Inconvenient Liquidity*

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Abstract

This paper develops a theory of positive and negative liquidity premia. Liquid assets facilitate decentralized exchange, but their marginal nonpecuniary value declines as they become more abundant and turns negative above a threshold—liquidity becomes inconvenient at the margin despite its positive social value. With multiple assets, positive and negative liquidity premia can coexist. Equilibrium exhibits ex post portfolio heterogeneity, with liquidity-constrained and unconstrained agents coexisting endogenously. A quantitative exercise generates liquidity premia of empirically plausible magnitudes and signs for long-maturity U.S. Treasuries. Finally, the socially optimal real return on money exceeds the Friedman-rule rate.

JEL Classification: E40, E50

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"The ultimate goal of a positive science is the development of a 'theory' or 'hypothesis' that yields valid and meaningful (i.e., not truistic) predictions about phenomena not yet observed."

— Milton Friedman, The Methodology of Positive Economics (1953), p. 7.

1 Introduction

Can an asset's moneyness *increase* its rate of return? Conventional wisdom says it cannot. Moneyness provides a positive nonpecuniary return—commonly referred to as a *liquidity premium* or *convenience yield* (we use the two terms interchangeably throughout the paper)—that reduces the pecuniary return investors require to hold assets.[1] Yet recent evidence suggests that this prediction does not always hold. For instance, convenience yields on Treasuries—highly liquid, money-like assets—are often negative (e.g., Fleckenstein and Longstaff, 2024). In this paper, we explain how an asset that facilitates exchange can nevertheless command a higher rate of return than an otherwise identical illiquid asset by showing that negative liquidity premia are a previously unrecognized implication of a canonical microfounded model of asset liquidity.[2]

To do so, we adopt an overlapping-generations version of the model of monetary exchange developed by Lagos and Wright (2003, 2005), hereafter LW, in which liquid assets take the form of Lucas (1978) trees. Assets are traded sequentially in a centralized market and a decentralized market with pairwise meetings and generalized Nash bargaining. While we retain the preferences, technologies, information structure, and trading environment of LW, the overlapping-generations structure allows us to characterize equilibria analytically over the entire parameter space, including regions where LW loses its analytical tractability. This characterization shows that liquidity premia become negative over part of that space. On the normative side, our model overturns the conventional policy prescriptions for the optimal provision of liquidity.

The key mechanism behind this result is that liquidity not only expands the set of feasible trades by mitigating agents' lack of commitment, but it also affects the division of surplus in decentralized exchange. In bilateral meetings, an agent facing a tighter liquidity constraint values concessions from her trading partner more. It is as if the agent were more patient in

[1] Krishnamurthy and Ma (2025, Section 3.3) describe the word "convenience" as a catchall word because a convenience yield can have several subcomponents, including liquidity, collateral, or safety.

[2] The pricing puzzle studied in this paper can be viewed as the mirror image of the classic rate-of-return dominance puzzle, according to which money earns a lower rate of return than otherwise similar assets. Recent contributions on this puzzle include Zhu and Wallace (2007), Lagos (2013), and Hu and Rocheteau (2015).

bargaining, allowing her to extract a larger share of the gains from trade.[3] As asset holdings increase, liquidity constraints relax and the share of the gains from trade accruing to agents with liquidity needs shrinks. Beyond some point, further increases in asset holdings make the marginal value of liquidity negative. It follows that when the asset supply is sufficiently large, market clearing can only be achieved if some agents hold asset positions for which their marginal nonpecuniary return is negative.

A negative liquidity premium means that the asset is priced below its fundamental value in the absence of liquidity considerations. So why don't agents take advantage of the asset's cheapness by purchasing arbitrarily large quantities? In our model, the same lack of commitment that gives rise to a demand for liquidity also limits arbitrage. Specifically, agents' asset purchases when young are constrained by their finite initial endowment, which we interpret as their *arbitrage capacity*. This limit to arbitrage is not the source of the negative liquidity premia in our model; rather, it allows the associated price discount to persist in equilibrium. Throughout the paper, we vary this arbitrage capacity and characterize equilibrium outcomes as it increases, including in the limit as it tends to infinity.

We start our general equilibrium analysis with a one-asset economy because it isolates the mechanism generating negative liquidity premia while facilitating comparisons with the existing literature. We define a *counterfactual* liquidity premium as the difference between the equilibrium return on a liquid asset and the return on the same asset in an otherwise identical economy in which it is not accepted in decentralized exchange. In accordance with the empirical literature (e.g., Krishnamurthy and Vissing-Jorgensen, 2012), our model identifies the supply of liquid assets as a central determinant of liquidity premia. When asset supply is low, the asset is priced above its value in the counterfactual economy (where the asset is illiquid), implying a positive liquidity premium. As the asset supply rises above some threshold, its real rate of return exceeds the counterfactual return, implying a negative liquidity premium. For a sufficiently large asset supply, the liquidity premium is zero, as the asset no longer provides liquidity services at the margin. Thus, when liquidity premia turn negative, the relationship between premia and asset supply is nonmonotone. Finally, as agents' arbitrage capacity (their initial endowment) increases, negative liquidity premia become smaller in absolute value but arise over a wider range of asset supplies, making them, paradoxically, more pervasive across the parameter space.

In the second part of the paper, we introduce multiple assets, some liquid and others illiquid, allowing us to define a *cross-sectional* liquidity premium consistent with empiri-

[3] The idea that illiquidity can have commitment value is reminiscent of Myers and Rajan (1998), although our mechanism is different and operates through the effect of liquidity constraints on bargaining outcomes.

cal measures. The main insight from the one-asset economy carries over: liquidity premia become negative once the supply of liquid assets exceeds a threshold. Unlike the one-asset economy, however, the cross-sectional liquidity premium remains negative even as the supply of the liquid asset becomes large. Thus, the option of investing in an illiquid asset expands the region of the parameter space where negative liquidity premia emerge. We calibrate this version of the model to the market for long-maturity Treasuries and show that it can account for the magnitude and sign of liquidity premia in the data.

Negative liquidity premia have novel implications beyond asset pricing. For a range of asset supplies consistent with negative liquidity premia, the endogenous distribution of asset portfolios is non-degenerate even though agents are ex ante identical and face no idiosyncratic risk when young. An endogenous fraction of agents remains liquidity constrained, while the remainder accumulates enough liquid assets to become unconstrained. Liquidity-constrained agents avoid the loss in surplus associated with abundant liquidity, whereas unconstrained agents exploit the discount at which liquid assets trade relative to otherwise identical illiquid assets. We think of the former as liquidity traders and the latter as arbitrageurs, with the endogenous composition of the two determined in equilibrium.

We also establish the role of negative liquidity premia in overturning a core policy prescription in monetary economics, the Friedman (1969) rule. In a version of the model with fiat money that pays a nominal interest rate financed by lump-sum taxes, the Friedman rule is suboptimal because negative liquidity premia arise while agents remain liquidity constrained. Hence, driving liquidity premia to zero—the Friedman rule recommendation—does not suffice to maximize welfare. The optimal policy balances the gain from compensating for the negative liquidity premium against the resulting intertemporal distortion, implying a rate of return on money *above* the Friedman-rule rate. The level that is optimal depends on agents' arbitrage capacity as well as trading frictions. Only in the limit as the endowment of the young becomes unbounded is the Friedman rule restored, in which case it implements the first-best allocation.

Finally, negative liquidity premia reshape liquidity-based yield curves. In the presence of multiple liquid assets that differ in their degree of liquidity, measured by their probability of acceptance in pairwise meetings, asset supplies and acceptability jointly determine the ordering of asset prices. Greater acceptability raises asset prices when liquidity premia are positive. The opposite occurs once liquidity premia become negative, so that more acceptable assets trade at lower prices. We also establish that positive and negative liquidity premia can coexist in equilibrium—some liquid assets are more expensive than illiquid ones while others are cheaper—in accordance with the evidence from U.S. Treasuries discussed below. The

framework also extends naturally to assets with different maturities, generating predictions for how the asset-supply threshold for negative liquidity premia varies with maturity and the size of the investor clientele for each maturity.

1.1 Empirical evidence on convenience yields

The empirical literature on convenience yields documents two robust regularities: convenience yields depend on asset supply and monetary conditions, and they are sometimes negative.[4] Convenience yields are typically measured as yield spreads between securities with similar cash flows but different liquidity properties. Examples include the spread between AAA corporate bonds and U.S. Treasury securities (Krishnamurthy and Vissing-Jorgensen, 2012), the difference between the rate on a collateralized repurchase agreement and that on a Treasury bill of the same maturity (Nagel, 2016), and the swap spread—the difference between the fixed rate on an interest-rate swap and the yield on a maturity-matched Treasury security (Fleckenstein and Longstaff, 2024; Jiang et al., 2025). These spreads provide empirical measures of the return investors are willing to forgo in exchange for liquidity services.[5]

Starting with Krishnamurthy and Vissing-Jorgensen (2012), empirical evidence points to asset supply and monetary policy as key determinants of convenience yields. Jiang et al. (2025) document a negative relationship between Treasury supply and convenience yields and argue, using an instrumental-variables strategy, that the relationship is causal. Nagel (2016), by contrast, emphasizes the role of the opportunity cost of money.

A central stylized fact documented by Fleckenstein and Longstaff (2024) is that Treasury convenience premia are frequently negative:

> "Many Treasury bills, notes, and bonds have repeatedly traded at significant discounts to their intrinsic fair values for extended periods throughout the entire sample period. (...) Negative convenience premia are frequently observed well before the beginning of the financial crisis."

He et al. (2022) document a particularly striking instance during the COVID-19 crisis, which they refer to as "Treasury inconvenience yields." Jiang et al. (2025) show that convenience yields on U.S. Treasuries are negative across several maturities; for example, the

[4] Krishnamurthy and Ma (2025) provide a review of the demand and supply of convenience assets, while Fleckenstein and Longstaff (2024) focus on the stylized facts regarding Treasury convenience premia.

[5] Consistent with this interpretation, Du et al. (2018, 2026) use deviations from covered interest parity to quantify the convenience yields of the U.S. dollar and U.S. Treasuries relative to foreign currencies and bonds.

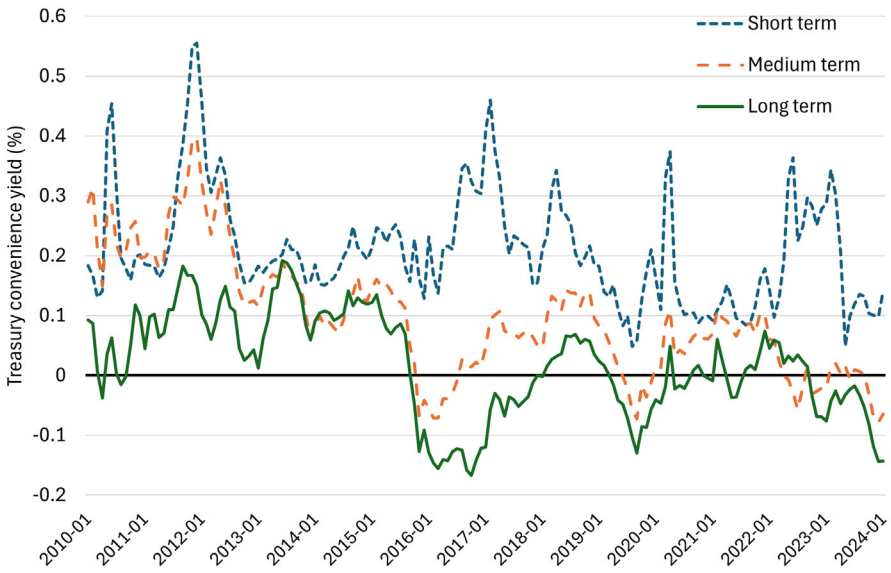


Figure 1: Treasury convenience yields measured as the difference between the swap rate and the maturity-matched Treasury yield. Short-term maturity: 1 to 3 years. Medium-term maturity: 3 to 7 years. Long-term maturity: 7 to 20 years. Adapted from Jiang et al. (2025).

10-year convenience yield reached approximately -0.25 percent in 2025. Figure 1 also shows that positive and negative premia can coexist across different securities. Du et al. (2026) document the decline in the convenience yield of U.S. Treasuries relative to foreign government bonds. Negative convenience yields are not confined to the United States: Fontalirand (2025) finds that five-year French government bonds have exhibited negative convenience yields since mid-2024.

1.2 Related literature

Our model is in the New Monetarist tradition and stays as close as possible to the LW model for the sake of comparison and analytical tractability. Versions of the LW model incorporating Lucas trees in order to generate positive liquidity premia include Geromichalos et al. (2007), Lagos (2010, 2011), and Rocheteau and Wright (2013), among others.[6] Most of this literature relies on "monotone" bargaining solutions, either take-it-or-leave-it offers or the Kalai (1977) solution. In contrast, we employ generalized Nash bargaining (with interior bargaining weights), which is the standard solution in the literature on decentralized markets,

[6] Extensions with multiple assets and heterogeneous liquidity properties study topics such as term premia, indirect liquidity, asset safety, and the relative illiquidity of TIPS (e.g., Geromichalos et al., 2016, 2023, 2024; Herrenbrueck and Geromichalos, 2017).

including LW, due to its natural axioms and strategic foundations. In the presence of liquidity constraints, the Nash solution exhibits a non-monotonicity that is key to generating negative liquidity premia.[7] This non-monotonicity property is not specific to the generalized Nash solution; it also arises under other standard axiomatic bargaining solutions, such as the Kalai and Smorodinsky (1975) solution.

The overlapping-generations structure follows Zhu (2008) and Nosal and Rocheteau (2006, p. 55). However, in our case, it is not introduced to study redistributive effects of inflation.[8] It is a formalization device used to characterize equilibria in regions where generalized Nash bargaining generates non-linear value functions and the standard LW framework becomes analytically intractable. Vayanos (1998) uses an overlapping-generations model to study the relationship between liquidity and asset prices and shows that this relationship can be counterintuitive: higher transaction costs can raise asset prices. The notion of liquidity and the underlying mechanism are fundamentally different, however: in Vayanos, illiquidity arises from the cost of trading the asset, whereas in our model liquidity refers to the asset's ability to relax constraints on decentralized trade.

The decentralized market in our model shares the same search-and-bargaining structure as the over-the-counter markets in Duffie et al. (2005) and the subsequent literature surveyed by Hugonnier et al. (2025). In these models, liquidity refers to the tradability of the asset itself, and liquidity yield spreads can be either positive or negative (Hugonnier et al., 2025, Section 4.5.3; see Longstaff, 2009, for an earlier example of this result). In our model, by contrast, liquidity refers to an asset's ability to facilitate transactions by relaxing liquidity constraints. This distinction is essential for generating moneyness and convenience yields.[9]

A related literature studies why swap rates can fall below maturity-matched Treasury yields. Klingler and Sundaresan (2019) argue that underfunded pension funds favor swaps to hedge their long-term liabilities because swaps provide exposure to long-term interest rates without requiring the purchase of Treasuries. Their demand to receive fixed rates pushes down swap rates, while limited dealer arbitrage allows the resulting negative swap spreads to persist. Jermann (2020) shows that Treasury holding costs, such as the cost of equity capital

[7] The importance of the non-monotonicity of Nash bargaining in models with liquidity constraints was first emphasized by Aruoba et al. (2007). This mechanism has since been used to generate endogenous credit cycles (Gu et al., 2013) and market shutdowns (Lebeau, 2020). In the axiomatic bargaining literature, obtaining monotonicity with respect to expansions of the bargaining set requires abandoning the scale invariance axiom, as in the proportional solution of Kalai (1977).

[8] Zhu's main result regarding the welfare benefits from positive money creation arises from strict concavity in old-age utility and is obtained under the assumption that the buyer's bargaining power is equal to or close to one. Instead, old-age utility is linear in our model and we consider bargaining powers bounded away from one.

[9] New Monetarist environments with OTC markets and liquidity constraints include Trejos and Wright (2016), Lagos and Zhang (2020), and Geromichalos and Jung (2024).

needed to satisfy leverage requirements, can generate negative swap spreads by limiting dealers' arbitrage between Treasuries and swaps. Du et al. (2023) emphasize Treasury supply: when dealers must absorb a large supply, Treasury yields rise relative to swap rates to compensate them for the opportunity cost of using scarce balance sheet capacity, measured by forgone profits from arbitrage exploiting deviations from covered interest parity. In contrast, the mechanism in this paper does not rely on costs of financing or holding the liquid asset. Instead, it is rooted in the moneyness of assets in decentralized markets in which agents bargain over prices. Our model also features limited arbitrage because finite endowments restrict positions in underpriced assets. This constraint affects the size of the price discount but is not its source: negative liquidity premia persist as arbitrage capacity becomes arbitrarily large.

2 Environment

Time is discrete, infinite, and indexed by $t \in \mathbb{N}_0$. Each period consists of a decentralized market (DM), in which agents are randomly matched in bilateral meetings, followed by a centralized market (CM). In the DM, each agent meets another agent with probability $\alpha \in (0, 1]$.

The economy is populated by a continuum of two-period-lived agents: a generation-t agent is born at the beginning of the CM in period $t-1$ and dies at the end of the CM in period t, as illustrated in Figure 2. Hence, two consecutive generations overlap only in the CM. The size of each generation is normalized to one.

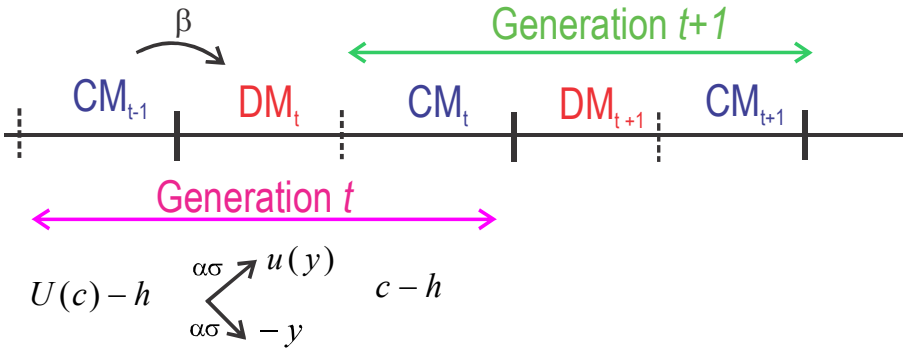


Figure 2: OLG structure of the model

In the DM, following Kiyotaki and Wright (1989), there is a perishable good available in $N \geq 3$ distinct varieties. Agents are specialized in both production and consumption. An agent of type $n \in \{1, ..., N\}$ produces good n and consumes good $n+1$ (modulo N). Agents

are uniformly distributed across types. Consequently, the probability that an agent values the good produced by a randomly matched counterparty is $\sigma \equiv 1/N$. Furthermore, there is no double coincidence of wants in pairwise meetings.

The utility from consuming y units of the DM good is $u(y)$, where $u \in C^\infty$ is strictly increasing and strictly concave, satisfies $u(0) = 0$, $u'(0) = +\infty$, and $u'(+\infty) = 0$. Producing $y \in \mathbb{R}^+$ units of DM output entails disutility y. We denote by $y^* \in (0, +\infty)$ the unique solution to $u'(y) = 1$. While y is a consumption good in our model, it is well understood that it could easily be reinterpreted as an input factor, or any dimension of a contract that generates gains from trade. Similarly, agents could be households, firms, or even financial institutions depending on the interpretation given to $u(\cdot)$ and y.

There is a single perishable good produced in the CM, which is taken as the numéraire. The technology to produce the CM good is linear in labor. Preferences in the CM when young are given by

$$\mathcal{U}_1(c, h) = U(c) - h,$$

where $U(\cdot)$ is strictly increasing, strictly concave, and twice continuously differentiable. It satisfies $U'(0) = +\infty$ and $U'(+\infty) = 0$. We denote by $c^* \in (0, +\infty)$ the unique solution to $U'(c) = 1$. Preferences in the CM when old are

$$\mathcal{U}_2(c, h) = c - h.$$

There is no discounting between the DM and CM stages within a period. The discount factor across periods is $\beta \equiv (1 + \rho)^{-1} \in (0, 1)$. The labor endowment of a young agent is $\bar{h}_1 > 0$, while that of an old agent is $\bar{h}_2 > 0$.[10]

Agents cannot commit to future actions and there is no public monitoring. Together with heterogeneous preferences and specialization in the DM, these frictions generate a role for liquid assets. For now, the only asset in the economy is a Lucas tree in fixed supply, $A > 0$, that pays a constant dividend, $d > 0$, in the CM. We denote by q_t the ex-dividend price of the tree in the CM at date t. The gross real rate of return of the Lucas tree is $R_{t+1} \equiv (q_{t+1}+d)/q_t$. Asset holdings are observable, and common knowledge, within bilateral meetings.

Discussion

A departure from LW is the introduction of overlapping generations. The role of the OLG structure is solely to characterize equilibria over the entire parameter space—including equi-

[10] New Monetarist models typically focus on equilibria where the labor supply constraint, $h \leq \bar{h}$, is non-binding in order to preserve the linearity of value functions and hence tractability. Tractable versions with binding labor constraints and ex post heterogeneity include Rocheteau et al. (2018, 2021).

libria with binding labor constraints and nonlinear value functions—while preserving analytical tractability. By contrast, the textbook Lagos and Wright (2003) model with Lucas trees and generalized Nash bargaining with an interior bargaining weight ceases to admit tractable equilibria once the asset supply exceeds a threshold.

The key friction generating a demand for liquid assets is the absence of commitment, which is imposed consistently throughout the model: agents cannot commit to *any* future action. In particular, they cannot commit to repay debts when old, regardless of whether those debts were incurred in the DM or in the CM when young.[11] Agents cannot be forced by other agents to work, nor can their assets be confiscated to settle private obligations. As a result, borrowing is incentive-infeasible both in the CM and DM. We distinguish this restriction from the government's power to collect a uniform tax in the CM, introduced in Section 4. We will investigate the robustness of our results to this assumption in several ways. We will increase the endowment of the young, $\bar{h}_1$, to arbitrarily high levels, thereby allowing agents to invest in arbitrarily large portfolios. We will also discuss the effect of an exogenous credit limit in DM trades.

Another consequence of lack of commitment is that agents cannot commit ex ante not to spend a fraction of their asset holdings. Later in the paper, we introduce illiquid assets, which allow agents to allocate part of their wealth to an asset that cannot be used in the DM, effectively committing them not to spend that portion of their wealth in the DM. Relatedly, we maintain the standard assumption in the literature (e.g., Lagos and Wright, 2005) that negotiations in pairwise meetings occur under complete information. Our online Appendix B5 studies a version of our model where agents can segregate some of their assets in a fraction $1-\lambda$ of the meetings.[12] Our results continue to hold for any $\lambda > 0$, i.e., provided that asset holdings cannot be concealed or segregated in all meetings.

We formalize moneyness parsimoniously through the asset's role as a means of payment in decentralized exchange. We do not explicitly model the use of assets as collateral in debt contracts, although the two roles are closely related, because doing so would require additional institutions, such as third-party custody and enforcement mechanisms to ensure the restitution of pledged collateral. Moreover, allowing the asset to serve as collateral would enable agents to borrow in the CM to finance their asset purchases, thereby introducing

[11] Our assumption of a generalized lack of commitment differs from Lagos and Wright (2005, fn. 2), who write: "Of course, (...) we can allow the exchange of intertemporal claims across centralized markets. Since agents are homogeneous in this market, such claims will not trade, but we can still price assets like real or nominal bonds." We show below that in our model, however, homogeneous agents can optimally choose different portfolios in the centralized market, despite being ex ante identical, for part of the parameter space where liquidity premia are negative.

[12] For models where agents can segregate a fraction of their asset holdings in all meetings, see, e.g., Berentsen and Rocheteau (2003) or Lebeau (2020).

the possibility of rehypothecation. In our Online Appendix B6, we establish formally that collateralized borrowing and lending with rehypothecation do not expand aggregate liquidity or arbitrage capacity, leaving equilibrium allocations and asset prices unchanged.

3 Equilibrium

In this section, we characterize the equilibria of the one-asset economy. Focusing first on a single asset allows us to isolate the mechanism leading to negative liquidity premia and facilitates comparisons with the existing literature. We define *counterfactual* liquidity premia by comparing the equilibrium rate of return in the economy where the asset is *liquid* and can be traded in pairwise meetings in the DM, where liquidity needs arise, with that in an otherwise identical economy in which the asset is *illiquid* and cannot be used in decentralized exchange. Section 5 extends the analysis to multiple assets. Throughout, we restrict attention to steady-state equilibria in which the asset's rate of return, R, is constant over time.

3.1 Value functions and bargaining

We characterize the value functions by working backward through the agent's life cycle, starting with the old agent in the CM, followed by the old agent in the DM, and finally the young agent in the CM. In the process, we characterize the outcome of negotiations in the DM.

CM value function when old Agents who lack commitment enter the CM when old with no outstanding debt obligations. Their wealth consists solely of Lucas trees. Let $W_o(a)$ denote the value function of an old agent entering the CM with a units of the asset, where the asset is valued at its cum-dividend price in the CM (and hence expressed in units of the numeraire). It solves

$$W_o(a) = \max_{c,h} \{c - h\} \quad \text{s.t.} \quad c = a + h, \;\; c \geq 0, \; h \in [0, \bar{h}_2]. \tag{1}$$

Substituting the budget constraint into the objective yields $W_o(a) = a$. This linearity keeps the bargaining problem tractable, to which we now turn.

DM bargaining In a pairwise meeting in the DM, a buyer and a seller negotiate the terms of trade, $(y, p) \in \mathbb{R}^2_+$, where y is the quantity of goods produced by the seller for the buyer and p is the payment expressed in the CM good. If the buyer starts the DM with a

units of asset, her surplus from trade is $u(y) + W_o(a-p) - W_o(a) = u(y) - p$. Similarly, the seller's surplus is $-y + p$. The negotiation is subject to a liquidity constraint: due to lack of commitment, the buyer cannot leave the meeting with negative wealth and hence her payment cannot exceed her asset holdings. The outcome of the negotiation is given by the generalized Nash solution,

$$\max_{y,p} \left[u(y) - p\right]^{\theta} \left[-y + p\right]^{1-\theta} \quad \text{s.t.} \quad p \leq a,$$

where $\theta \in (0,1)$ is the buyer's bargaining power, which is assumed to be interior, and a is asset holdings expressed in terms of the numeraire. If $p \leq a$ is not binding, $y = y^*$ and $p = p^* \equiv \theta y^* + (1-\theta)u(y^*)$. Otherwise,

$$a = p = w(y) \equiv \frac{\theta u'(y) y + (1-\theta)u(y)}{\theta u'(y) + 1 - \theta}. \tag{2}$$

For all $y \in [0, y^*]$, $w(y)$ is a strictly increasing function. Thus, $y(a)$, implicitly defined by $w(y) = a$, is increasing in a.

We define the buyer's surplus from a DM trade as $\Delta^b(a) \equiv u\left[y(a)\right] - p(a)$, which can be reexpressed as

$$\Delta^b(a) = \begin{cases} u \circ w^{-1}(a) - a & \text{if } a < p^* \\ \theta\left[u(y^*) - y^*\right] & \text{if } a \geq p^*. \end{cases} \tag{3}$$

The function $\Delta^b(a)$ has the following critical property: for all $\theta \in (0,1)$,

$$\Delta^{b\prime}(p^{*-}) < 0 = \Delta^{b\prime}(p^{*+}).$$

Thus, $\Delta^b(a)$ is not differentiable at $a = p^*$, but it is differentiable everywhere else. As a approaches p^* from below, Δ^b decreases. Thus, near the efficient allocation y^*, relaxing the liquidity constraint *lowers* the buyer's surplus despite increasing total trade surplus. As a approaches p^* from above, Δ^b stays constant. The kink at $a = p^*$ generates a non-convexity that will have important implications in the following.

In order to explain the non-monotonicity of $\Delta^b(a)$, we express the buyer's surplus as the product of two terms, the joint surplus, $u(y) - y$, and the buyer's share in that surplus, $\Theta(y)$, i.e.,

$$\Delta^b = \Theta(y)\left[u(y) - y\right] \text{ where } \Theta(y) \equiv \frac{\theta u'(y)}{\theta u'(y) + 1 - \theta}.$$

Relaxing the liquidity constraint increases y and therefore the total trade surplus. However, it also lowers the buyer's endogenous bargaining share, $\Theta(y)$. Near the efficient allocation y^*, the increase in total surplus is second-order, whereas the decline in the buyer's bargaining

share is first-order, i.e., $\partial\Delta^b/\partial y = \Theta'(y^*)\left[u(y^*) - y^*\right] < 0$. Thus, for all $\theta \in (0,1)$, the asset holdings that maximize $\Delta^b(a)$ are lower than p^*.[13]

What is the economic intuition for $\Theta'(y) < 0$? The mechanism is analogous to the effect of payroll taxes in models of the labor market where wages are negotiated according to the Nash solution. A payroll tax creates a wedge between the firm's marginal cost of increasing the wage and the worker's marginal benefit, thereby altering the division of the surplus even though the bargaining weights themselves remain unchanged. Likewise, a binding liquidity constraint creates a wedge between the buyer's and seller's marginal valuations of transferring an additional unit of the liquid asset. As a result, the endogenous division of the surplus changes even though the primitive bargaining weights remain unchanged. In the online Appendix B2, we provide strategic foundations for these bargaining shares using an extensive-form alternating-offers bargaining game with exogenous risks of breakdown.[14]

DM value functions The value of an agent entering the DM with a units of asset is

$$\begin{aligned} V(a) &= \alpha\sigma\left\{u\left[y(a)\right] + W_o\left[a - p(a)\right]\right\} \\ &\quad + \alpha\sigma\int -y(a') + W_o\left[a + p(a')\right] dF(a') + (1 - 2\alpha\sigma)W_o(a), \end{aligned} \tag{4}$$

where $F(a')$ is the equilibrium distribution of asset holdings across agents in the DM. We will show later that this distribution can be nondegenerate. The first term on the right side represents the event where the agent is a buyer in a pairwise meeting, with probability $\alpha\sigma$. The second term is when the agent is the seller in the match. Using that $W_o(a) = a$, (4) can be simplified as

$$V(a) = \alpha\sigma\Delta^b(a) + \alpha\sigma\Delta^s + a, \tag{5}$$

where Δ^s is the seller's expected surplus in equilibrium. In the following, in order to characterize young agents' asset demands, we impose restrictions on $\Delta^b(a)$ that will be satisfied, for instance, if u is of the CRRA form (as shown in the online Appendix B1).[15]

Assumption 1a. *The function $\Delta^b(a)$ restricted to $[0, p^*]$ is strictly unimodal (and hence*

[13] The non-monotonicity of $\Delta^b(a)$ is not specific to the generalized Nash solution. It also arises under other standard axiomatic bargaining solutions, such as the Kalai-Smorodinsky solution. In the axiomatic bargaining literature, obtaining monotonicity with respect to expansions of the bargaining set requires abandoning the axiom of scale invariance (Kalai, 1977).

[14] The extensive-form game has an infinite number of rounds. Since it takes place *within a period*, we could assume instead a large but finite number of rounds. The solution would approximate that of the infinite-horizon game. In online Appendix B3, we describe a simple two-round game, as in Zhu (2019). While it does not provide strategic foundations for the Nash solution, it illustrates that the non-monotonicity of the buyer's surplus arises in many bargaining games with alternating offers.

[15] This assumption is not necessary to analyze the model (see, e.g., Gu and Wright, 2016).

strictly quasi-concave) and it satisfies $\Delta^{b\prime}(0) = +\infty$. Its maximum is reached at $\tilde{a} \in (0, p^*)$, where $\tilde{a}$ solves $\Delta^{b\prime}(\tilde{a}) = 0$.

CM value function when young We now turn to the problem of an agent in the CM when young. It is given by

$$W_y = \max_{c\geq 0, h\geq 0, a\geq 0} \{U(c) - h + \beta V(a)\} \quad \text{s.t.} \quad c + \frac{a}{R} = h \quad \text{and } h \leq \bar{h}_1.$$

Young agents, who cannot borrow against their future income, use their current labor to finance consumption and asset purchases. Replacing $V(a)$ with its expression in (5), we obtain

$$W_y = \max_{a\in\left[0, R\bar{h}_1\right]} \Gamma(a) + \beta\alpha\sigma\Delta^s, \tag{6}$$

where the function Γ is defined as

$$\Gamma(a) \equiv \max_{c\leq \bar{h}_1 - a/R} \left\{ U(c) - c + \beta \left[-\left(\frac{1-\beta R}{\beta R}\right) a + \alpha\sigma\Delta^b(a) \right] \right\} \quad \forall a \in \left[0, R\bar{h}_1\right]. \tag{7}$$

The first two terms between brackets correspond to the surplus from CM consumption while the remaining terms represent the net surplus from DM consumption. The constraint on c creates a potential trade-off between these two surpluses.

We make the following assumption on primitives that ensures that the constraint $c \leq \bar{h}_1 - a/R$ is slack when agents are liquidity constrained, $a \leq p^*$. We assume that young agents have enough endowment to finance both c^* and p^*. It makes the model more directly comparable to the LW model.

Assumption 2. *The labor endowment when young satisfies* $\bar{h}_1 \geq c^* + p^*$.

3.2 Asset demand

We define the individual asset demand correspondence as

$$\text{A}^*(R) \equiv \arg \max_{a\in\left[0, R\bar{h}_1\right]} \Gamma(a).$$

In order to tackle the non-differentiability and non-convexity at p^*, we divide the domain for Γ into two sub-intervals and we define:

$$g_1 \equiv \Gamma|_{[0,p^*]} \quad \text{and} \quad g_2 \equiv \Gamma|_{[p^*, R\bar{h}_1]}.$$

The function g_1 is equal to Γ restricted to the domain $[0, p^*]$ while g_2 is restricted to $[p^*, R\bar{h}_1]$. We establish in the Appendix (Lemma 3) that the correspondence $\text{A}_1^*(R) \equiv$

$\arg\max_{a\in[0,p^*]} g_1(a;R)$ is upper hemicontinuous and we denote a generic element as $a_1^* \in \mathrm{A}_1^*$. Any selection from $\mathrm{A}_1^*(R)$ is increasing in R. If $R < \beta^{-1}$, then $a_1^* < \tilde{a}$. If $R = \beta^{-1}$, $a_1^* = \tilde{a}$. Finally, if $R > \beta^{-1}$, then $a_1^* > \tilde{a}$.

We also establish (Lemma 4) that g_2 is concave, and strictly concave when a exceeds a threshold. Thus, $\mathrm{A}_2^*(R) \equiv \arg\max_{a\in[p^*,R\bar{h}_1]} g_2(a;R)$ is upper hemicontinuous and convex-valued. We denote a generic element as $a_2^* \in \mathrm{A}_2^*$. Within the g_2 region, if $R < \beta^{-1}$, $a_2^* = p^*$. If $R = \beta^{-1}$, the agent is indifferent among all admissible values of a that sustain $c = c^*$. If $R > \beta^{-1}$, it is optimal to accumulate assets beyond $R\left(\bar{h}_1 - c^*\right)$, which leads to consumption below c^*. These cases are represented in Figure 3.

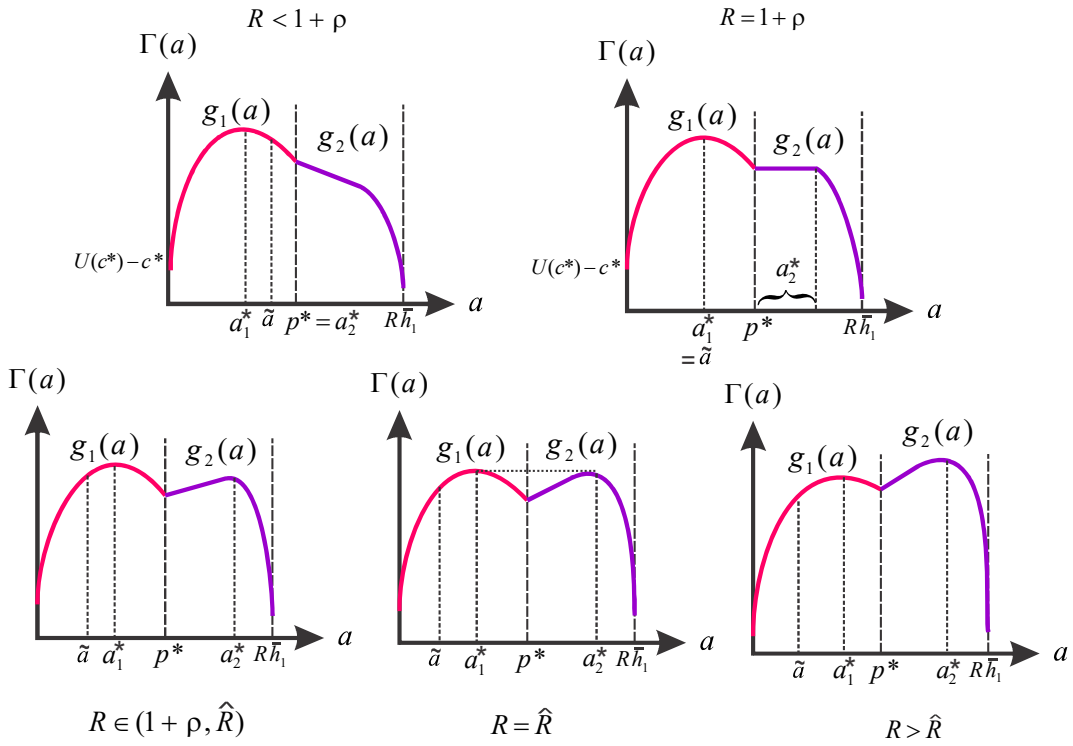


Figure 3: Functions $\Gamma(a)$, $g_1(a)$, and $g_2(a)$

Agents' asset demand corresponds to the maximization of $\Gamma(a)$ over the whole interval, $\left[0, R\bar{h}_1\right]$, which is analyzed in the next proposition. It shows that the agent's portfolio problem exhibits two distinct regimes. In one regime, agents hold assets primarily for liquidity purposes. In the other, agents hold assets beyond their liquidity needs despite the lower DM surplus associated with abundant liquidity.

Proposition 1 ***(Asset demand.)*** *There exists a unique* $\hat{R} > \beta^{-1}$ *solution to*

$$\max_{a\in[0,p^*]} g_1(a;\hat{R}) = \max_{a\in[p^*,\hat{R}\bar{h}_1]} g_2(a;\hat{R}). \tag{8}$$

The threshold, $\hat{R}$, is decreasing in $\bar{h}_1$ and increasing in $\alpha\sigma$. The asset demand from a young agent is such that:

1. *For all $R < \hat{R}$, $\mathrm{A}^*(R) = \mathrm{A}_1^*(R)$ where any $a_1^* \in \mathrm{A}_1^*(R)$ is a solution to*

$$\alpha\sigma\Delta^{b\prime}(a_1^*) = \frac{1-\beta R}{\beta R}. \tag{9}$$

2. *For all $R > \hat{R}$, $\mathrm{A}^*(R) = \mathrm{A}_2^*(R) = \{a_2^*\}$ where $a_2^* \in \left(R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right)$ solves*

$$U'\left(\bar{h}_1 - \frac{a_2^*}{R}\right) = \beta R. \tag{10}$$

3. *If $R = \hat{R}$, $\mathrm{A}^*(R) = \mathrm{A}_1^*(R) \cup \mathrm{A}_2^*(R)$.*

4. *Finally, any selection from A^* is strictly increasing in R and grows unbounded as $R \to +\infty$.*

There are two motives for holding assets: a *liquidity* motive, whereby agents accumulate assets to relax their liquidity constraint in the DM, and an *intertemporal-substitution* motive, whereby young agents accumulate assets to shift consumption from the current CM to the future CM. When $R < 1+\rho$, intertemporal substitution discourages asset accumulation, while the liquidity motive is operative. Once $R > 1 + \rho$, the intertemporal-substitution motive becomes positive while the marginal liquidity motive becomes negative. Agents continue to hold the asset for liquidity purposes, but additional liquidity now lowers their trade surplus.

There is a threshold for R, denoted $\hat{R} > 1 + \rho$, below which agents choose portfolios that leave them liquidity constrained in the DM, $a < p^*$. In that case, the optimal choice is characterized by (9), where the right side captures the opportunity cost of holding assets, measured by the spread $s \equiv (\rho - r)/R$, with $r = R - 1$. The left side captures the liquidity value, or convenience yield, of the marginal unit of the asset. When $s = 0$, the cost of holding the asset is zero and therefore the liquidity value at the margin must also be zero, which occurs at $a_1^* = \tilde{a}$ (top-right panel of Figure 3). When $s > 0$, agents economize on their holdings of liquid assets, so that $a_1^* < \tilde{a}$ (top-left panel of Figure 3). Finally, when $s < 0$, agents are willing to hold more than $\tilde{a}$ because the asset's pecuniary return compensates for its negative liquidity value at the margin (bottom-left panel of Figure 3).

If $R > \hat{R}$, liquidity no longer affects the marginal value of asset holdings, so the choice of assets is driven solely by intertemporal substitution between the current and future CM, as shown by the standard Euler equation, (10). In this region, agents hold more than p^*,

and liquidity constraints are slack. However, the labor supply constraint, $h \leq \bar{h}_1$, is binding. This case is illustrated in the bottom right panel of Figure 3.

At $R = \hat{R}$, agents are generically indifferent between two optimal choices, $a_1^* < p^*$ and $a_2^* > p^*$, as shown in the bottom middle panel of Figure 3. Although holding a_2^* yields a lower DM trade surplus—since $\Delta^b(a_2^*) = \Delta^b(p^*) < \Delta^b(a_1^*)$—it generates a higher net intertemporal surplus across CMs, which exactly offsets the lower expected DM surplus at the switching return.

3.3 Asset market equilibrium

We now turn to the determination of the equilibrium value for R. In order to allow for asymmetric equilibria, so that agents can make different choices of asset holdings when they are indifferent, we define the aggregate asset demand as the convex hull of $\mathrm{A}^*(R)$,

$$\mathrm{A}^d(R) = \mathrm{conv}[\mathrm{A}^*(R)].$$

From Proposition 1, any selection from $\mathrm{A}^d(R)$ is increasing in R. Recall that the demand for assets has been expressed in terms of the CM good. Hence, the asset supply must also be expressed in the numeraire, which we obtain by multiplying the quantity of asset, A, by its cum-dividend price in the CM, $q + d$, where $q = d/(R-1)$. It follows that the asset supply expressed in the numeraire is

$$A(q+d) = A\left(\frac{dR}{R-1}\right). \tag{11}$$

From the right side of (11), the real asset supply is decreasing in R. Market clearing requires that

$$Ad\left(\frac{R}{R-1}\right) \in \mathrm{A}^d(R). \tag{12}$$

An equilibrium is an $R^* \in (1, +\infty)$ solution to (12).

The following proposition shows that as A increases, the economy transitions from a regime with binding liquidity constraints to a regime with heterogeneous asset holdings with both binding and slack liquidity constraints, and finally to a regime with slack liquidity constraints.

Proposition 2 ***(Existence and uniqueness of the equilibrium rate.)*** *An equilibrium exists, and the equilibrium gross real rate of return, R^*, is unique and weakly increasing in A. There exist $A_0 < A_1 < A_2$ such that the following is true:*

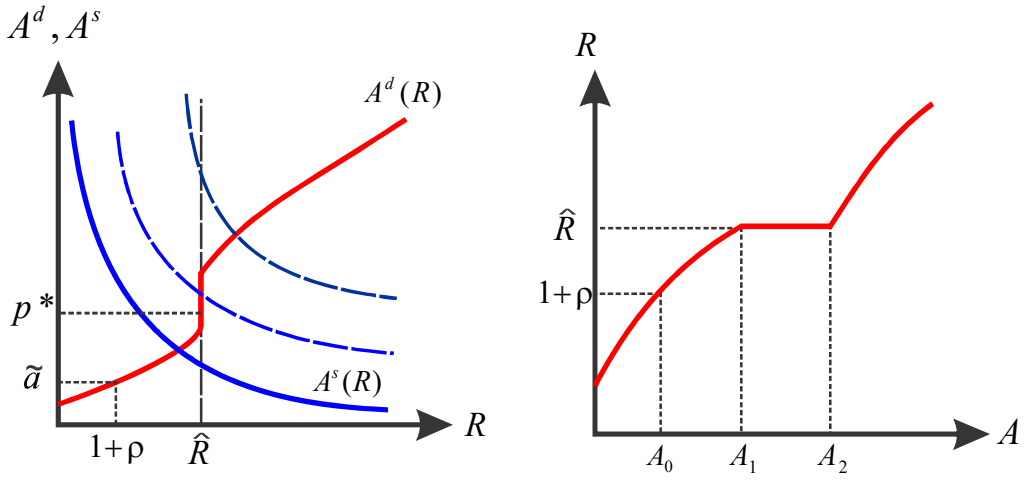


Figure 4: Left panel: asset demand and asset supply; Right panel: Equilibrium real interest rate as a function of A.

1. *For all $A < A_0$, $R^* < 1+\rho$ is the solution to*

$$R^* = \frac{1}{\beta\left[1+\alpha\sigma\Delta^{b\prime}(a_1^*)\right]} \quad \forall a_1^* \in \mathrm{A}_1^*(R^*), \tag{13}$$

   where $\alpha\sigma\Delta^{b\prime}(a_1^*) > 0$.

2. *For all $A \in (A_0, A_1)$, $R^* \in (1+\rho, \hat{R})$ solves (13) where $\alpha\sigma\Delta^{b\prime}(a_1^*) < 0$.*

3. *For all $A \in (A_1, A_2)$, $R^* = \hat{R} > 1+\rho$ where $\hat{R}$ solves (8). Generically, there is a nondegenerate distribution of asset holdings where a fraction of agents choose $a_1^* < p^*$ and a fraction choose $a_2^* > p^*$.*

4. *For all $A > A_2$, $R^* > \hat{R}$ solves*

$$R^* = \beta^{-1}U'\left(\bar{h}_1 - \frac{Ad}{R^*-1}\right). \tag{14}$$

Proposition 2 distinguishes four equilibrium regions. When A is below a threshold A_0, R^* is below the gross rate of time preference. In this case, agents economize on their asset holdings and choose levels below $\tilde{a}$, the amount that maximizes their trade surplus in the DM. The threshold, A_0, is the asset supply at which R^* is equal to $1+\rho$. As A rises above A_0, liquidity is so abundant that agents must hold more than $\tilde{a}$ for the market to clear. Hence, buyers operate on the downward-sloping portion of $\Delta^b(a)$ so that their DM trade surplus declines, i.e., $\Delta^{b\prime}(a) < 0$, and R^* exceeds $1+\rho$. The gross real rate of return, R^*, keeps increasing until A reaches some second threshold, A_1.

When A is between A_1 and A_2, R^* is constant and equal to $\hat{R} > 1 + \rho$. The equilibrium features a nondegenerate, generically two-point, distribution of asset holdings. Changes in asset supply are absorbed through changes in the distribution of portfolios rather than through movements in R^*. Indeed, an endogenous fraction, ω, of agents choose to hold $a_1^*(\hat{R}) < p^*$ while the remaining $1-\omega$ hold $a_2^*(\hat{R}) > p^*$. The heterogeneous portfolio allocation reflects agents' indifference between two portfolio choices: holding low liquidity balances to preserve DM surplus versus holding abundant liquidity to exploit the cheapness of the asset and hence its high return. The first group remains liquidity constrained in the DM—we think of them as *liquidity traders*—whereas the second group does not: they exploit the asset's cheapness, but their arbitrage positions remain constrained by their initial endowment when young—we think of them as endowment-constrained *arbitrageurs*. Finally, when A is larger than A_2, liquidity is sufficiently abundant to relax all constraints in the DM, in which case R^* is larger than $\hat{R}$.

3.4 Counterfactual liquidity premia

We define the liquidity premium by comparing the equilibrium return, R^*, with the equilibrium return, R^0, in an otherwise identical economy in which the asset is not traded in the DM, $\alpha = 0$. Since the asset's moneyness arises exclusively through decentralized exchange, this counterfactual shuts down the asset's nonpecuniary return while leaving all other features of the economy unchanged. Formally, the counterfactual liquidity premium is defined as

$$\mathcal{L} \equiv R^0 - R^*. \tag{15}$$

In order to determine R^0, the following lemma characterizes the aggregate asset demand in the absence of liquidity needs.

Lemma 1 *Suppose $\alpha = 0$. The asset demand correspondence is given by:*

$$\text{A}^d(R) = \begin{cases} \{0\} \\ [0, R(\bar{h}_1 - c^*)] \\ \left\{R\left[\bar{h}_1 - U'^{-1}(\beta R)\right]\right\} \end{cases} \quad \textit{if} \quad R \begin{cases} < \\ = \ \beta^{-1}. \\ > \end{cases} \tag{16}$$

If $\alpha = 0$, the only motive driving asset demand is intertemporal substitution. When $R < \beta^{-1}$, the asset return is too low to compensate agents for their rate of time preference, so young agents choose not to save. When $R = \beta^{-1}$, agents are indifferent over asset holdings as long as they can still finance c^*. Finally, when $R > \beta^{-1}$, the return on the asset is sufficiently high that agents are willing to reduce current consumption below c^* in order to accumulate additional assets. The asset demand correspondence is plotted in the left panel of Figure 5. The next proposition characterizes the value R^0 that clears the asset market.

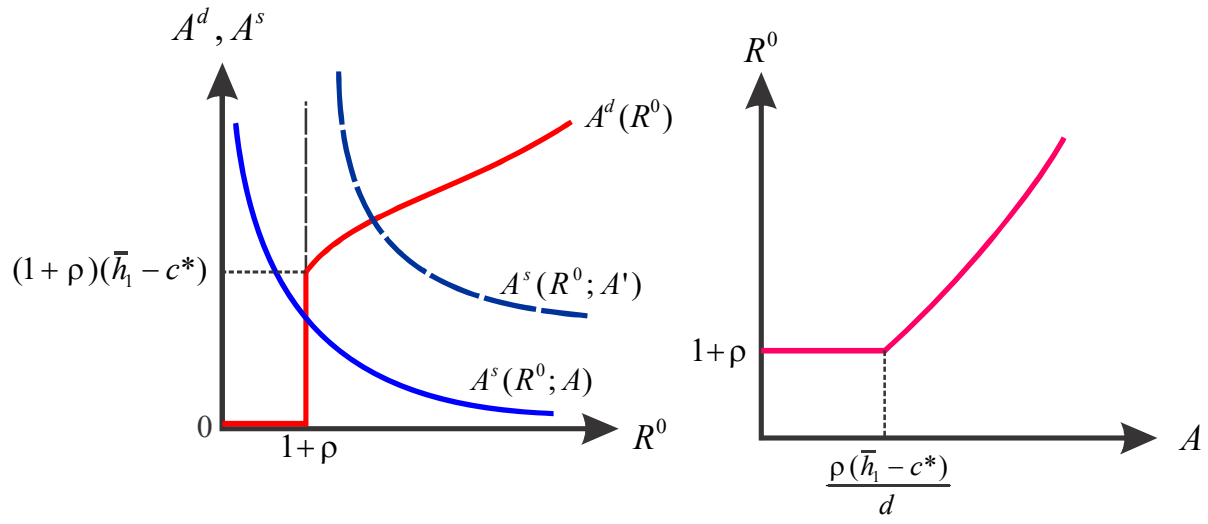


Figure 5: Equilibrium of the asset market in the absence of liquidity needs. Left panel: asset demand and asset supply (expressed in the numeraire); the aggregate asset supply is drawn for A and $A' > A$. Right panel: Equilibrium real interest rate as a function of A.

Proposition 3 ***(Equilibrium without liquidity needs.)*** *Consider the economy with* $\alpha = 0$. *There exists a unique equilibrium and it is such that:*

$$R^0 = \beta^{-1} \quad \text{if} \quad A \leq \breve{A}_1 \equiv \frac{\rho\left(\bar{h}_1 - c^*\right)}{d} \in (A_0, A_2) \tag{17}$$

$$= \beta^{-1} U'\left(\bar{h}_1 - \frac{dA}{R^0 - 1}\right) \quad \text{if} \quad A > \breve{A}_1. \tag{18}$$

Proposition 3 highlights two distinct regimes in the economy without liquidity needs. When the supply of the asset is sufficiently low, specifically when $dA \leq \rho\left(\bar{h}_1 - c^*\right)$, agents can simultaneously finance c^* and absorb the aggregate asset supply while earning the rate of time preference. In that case, the equilibrium satisfies $R^0 = \beta^{-1}$. When the asset supply becomes sufficiently large, however, young agents cannot simultaneously finance both c^* and the acquisition of the aggregate asset supply at $R^0 = \beta^{-1}$ because of the labor endowment constraint. As a result, the equilibrium real interest rate rises above the rate of time preference in order to induce agents to reduce current consumption and absorb the additional asset supply. Consequently, R^0 becomes increasing in A. The right panel of Figure 5 plots R^0 as a function of the quantity of assets in the economy.

We can turn to the implications of our model for liquidity premia. The following corollary to Propositions 2 and 3 characterizes the conditions under which liquidity premia are positive or negative.

Corollary 1 ***(Positive and negative liquidity premia.)*** *Consider the economy with positive liquidity needs, $\alpha > 0$.*

1. *If $A \leq A_0$, then $\mathcal{L} \geq 0$, i.e., $R^* \leq R^0 = 1 + \rho$;*
2. *If $A \in (A_0, A_2)$, then $\mathcal{L} < 0$, i.e., $R^* > R^0$;*
3. *If $A \geq A_2$, then $\mathcal{L} = 0$, i.e., $R^* = R^0$.*

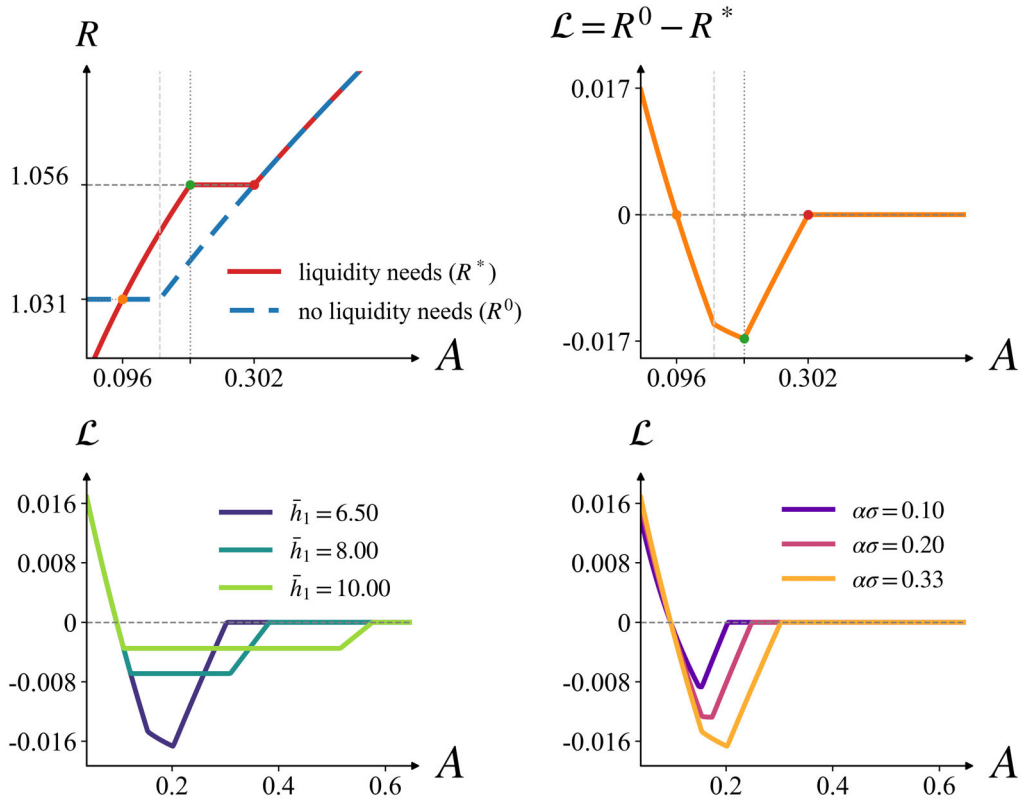


Figure 6: Top-left panel: Asset's gross real rate of return in the economy without liquidity needs (R^0) and in the economy with liquidity needs (R^*) as a function of asset supply. Top-right panel: Liquidity premium as a function of asset supply. Bottom-left panel: Liquidity premium for different values of $\bar{h}_1$. Bottom-right panel: Liquidity premium for different values of $\alpha\sigma$. Unless otherwise indicated, the benchmark functional forms and parameter values are $U = 5\log(c)$, $u(y) = y^{0.4}/0.4$, $\beta = 0.97$, $\theta = 0.7$, $\alpha\sigma = 0.33$, $d = 0.3$, and $\bar{h}_1 = 6.5$.

In the top left panel of Figure 6, we plot R^* and R^0 where the former is depicted by the red curve and the latter by the blue dashed curve. In the top right panel, we plot the liquidity premium, $R^0 - R^*$. When $A < A_0$, we have $R^* < R^0$, reflecting a positive liquidity premium: agents facing binding liquidity constraints in the DM value the asset

at the margin for its moneyness. This nonpecuniary return declines as the asset supply increases. For $A \in (A_0, A_1)$, R^* exceeds R^0, implying a negative liquidity premium that becomes more negative as A increases. Although liquidity constraints remain binding in this region, their relaxation reduces buyers' trade surplus in the DM. The liquidity of the asset is an *inconvenient* attribute at the margin. For $A \in (A_1, A_2)$, R^* remains constant at $\hat{R}$ and continues to exceed R^0. Since R^0 rises with A once the labor constraint binds, the liquidity premium becomes less negative as A increases. For $A > A_2$, $R^* = R^0$: liquidity constraints are slack and the liquidity premium vanishes.

The bottom panels of Figure 6 show the comparative statics of the liquidity premium with respect to $\alpha\sigma$ and $\bar{h}_1$. The parameter α captures how the asset's marketability affects the size of the convenience yield. When $\mathcal{L} > 0$, the liquidity premium increases as the asset becomes easier to trade, that is, as $\alpha\sigma$ rises. When $\mathcal{L} < 0$, improvements in marketability make the liquidity premium even more negative. Next, we turn to the comparative statics with respect to $\bar{h}_1$.

Arbitrage opportunities When $R^* > R^0$, the asset is cheap in that it is priced below its value in the economy with no liquidity needs or, equivalently, an economy without DM. If agents had large resources, could they exploit this difference by buying a large quantity of the asset, well above their liquidity needs, in order to resell those assets in the next CM when old? In our model, lack of commitment prevents young agents from borrowing against their future income. However, this friction is mitigated—but not eliminated—by the endowment of the young, $\bar{h}_1$, that we interpret as their *arbitrage capacity*.[16] Indeed, a higher $\bar{h}_1$ increases agents' maximum amount of assets they can purchase when young. As shown in the bottom-left panel of Figure 6, negative liquidity premia arise regardless of the magnitude of $\bar{h}_1$. As $\bar{h}_1$ increases, liquidity premia become less negative, but the range of asset supplies, A, for which $\mathcal{L} < 0$ expands. It is only at the limit, when $\bar{h}_1 \to +\infty$, that the difference between R^0 and R^* is driven to 0.

Liquidity premia and credit How would access to credit in the DM affect liquidity premia? One could introduce an exogenous and enforceable within-period credit limit, ℓ, that can be used as payment in the DM and repaid in the subsequent CM. The old-age labor endowment is assumed large enough to make repayment feasible even if the buyer spends all her liquid assets and uses the full credit line. The DM trade surplus would then be equal

[16] Online Appendix B6 allows young agents to finance asset purchases through collateralized borrowing with rehypothecation. Under the collateral and settlement rules specified there, these arrangements leave real allocations and asset prices unchanged.

to $\Delta^b(\ell + a)$. If $\ell < \tilde{a}$, the analysis is qualitatively similar to that above, except that the threshold asset supply at which liquidity premia become negative is lower. If $\tilde{a} < \ell < p^*$, liquidity premia are never positive: they are negative when asset supply is low and zero when asset supply is sufficiently large. If $\ell \geq p^*$, assets play no liquidity role, and the economy coincides with the one in Proposition 3.

4 Money and the optimal provision of liquidity

We now examine the implications of negative liquidity premia for the value of fiat money and optimal monetary policy. We will show that the same mechanism that generates negative liquidity premia induces a socially optimal rate of return of money higher than the Friedman rule.

In the absence of policy intervention, fiat money corresponds to the case $d \to 0$. There exists an equilibrium where fiat money has a positive price solution to $\alpha\sigma\Delta^{b\prime}(Aq) = \rho$. For all $\rho > 0$, aggregate real money balances are such that $Aq < \tilde{a} < p^*$.

In order to study monetary policy, we assume that the government sets a nominal interest rate on money, $i_m \geq 0$.[17] Formally, $d = i_m q$ and $R = 1 + i_m$. The interest payment is financed by a lump-sum tax, τ, denominated in the CM good and imposed on old agents. We assume that the government can enforce this tax in the CM without enforcing private promises. Thus, the government's budget constraint is

$$Aqi_m = \tau. \tag{19}$$

Assumption 3 (Tax feasibility). *The endowments satisfy* $\bar{\iota} \equiv \bar{h}_2/\bar{h}_1 > \hat{R} - 1 \equiv \hat{\iota}$, where $\hat{R}$ solves (8).

We restrict monetary policy to $i_m \in [0, \bar{\iota})$. For every such policy, old agents can pay the lump-sum tax regardless of their money holdings. Indeed, the resource constraint of young agents and market clearing imply $Aq < \bar{h}_1$, since young agents consume $c > 0$. Hence, from (19), $\tau = Aqi_m \leq i_m\bar{h}_1 < \bar{h}_2$. In particular, the tax is feasible even for an old agent who enters the CM with no money.

4.1 Liquidity premia and the rate of return of money

From now on, we impose the following restrictions on $\Delta^b(a)$:

[17] For a version of the Lagos-Wright model where money pays interest financed with lump-sum taxes, see Rocheteau and Nosal (2017, Chapter 6.2). This policy is equivalent to implementing a rate of return of money via the growth rate of the money supply.

Assumption 1b. *The function* $\Delta^b(a)$ *restricted to* $[0, p^*]$ *is strictly concave* and it satisfies $\Delta^{b\prime}(0) = +\infty$ and $\lim_{a\to 0} \Delta^{b\prime}(a)a = 0$.

These restrictions hold, for instance, if $u(y) = y^{1-\gamma}/(1-\gamma)$ and γ is not too large (see the online Appendix B1). We focus on symmetric equilibria where all agents choose the same real balances, a^*. Hence, the market-clearing condition is

$$a^* = Aq(1+i_m) \in \text{A}^*(1+i_m), \tag{20}$$

where $\text{A}^*(R)$ is the individual asset demand correspondence defined in Proposition 1. A steady-state, symmetric, monetary equilibrium is a $q > 0$ solution to (20) and a tax, τ, solution to (19), given $i_m \in [0, \bar{\iota})$.

Proposition 4 ***(Monetary equilibria with interest on currency.)*** *Under Assumption 3, for all* $i_m \in [0, \bar{\iota})$*, there exists a steady-state, symmetric, monetary equilibrium. The threshold* $\hat{\iota}$ *satisfies* $\hat{\iota} > \rho$*, and the following is true:*

1. ***Money as a medium of exchange****. For all* $i_m \in [0, \hat{\iota})$*,* $a^* = (1+i_m)qA < p^*$ *solves*

$$\alpha\sigma\Delta^{b\prime}(a^*) = \frac{\rho - i_m}{1+i_m}. \tag{21}$$

2. ***Money as a store of value****. For all* $i_m \in (\hat{\iota}, \bar{\iota})$*,* $a^* = (1+i_m)(\bar{h}_1 - c) > p^*$*, where* $c < c^*$ *solves*

$$U'(c) - 1 = \frac{i_m - \rho}{1+\rho}. \tag{22}$$

3. *If* $i_m = \hat{\iota}$*, then there are two symmetric monetary equilibria,* $a^* \in \{a_1^*, a_2^*\}$*, where* $a_1^* < p^*$ *is the solution to (21) and* $a_2^* = (1+i_m)(\bar{h}_1 - c) > p^*$*, where* $c < c^*$ *solves (22).*

There exists a threshold for the interest rate on money, $\hat{\iota}$, below which money is held at the margin for its liquidity services, and above which it is held at the margin for its store-of-value role. This threshold exceeds the rate of time preference, $\hat{\iota} > \rho$, which corresponds to the Friedman rule in the LW model.

Consider first the case where $i_m < \hat{\iota}$. The optimal choice of real balances is characterized by (21), where the left side is the liquidity return and the right side is the opportunity cost. When $i_m < \rho$ the opportunity cost is positive and agents economize on real money holdings, $a^* < \tilde{a}$. When $i_m = \rho$, holding money is costless and agents therefore choose $a^* = \tilde{a} < p^*$, which maximizes their DM trade surplus.

When $i_m \in (\rho, \hat{\imath})$, the monetary equilibrium is such that agents choose real balances in the interval $(\tilde{a}, p^*)$. Thus, it is possible to implement an interest rate on money above ρ.[18] Despite earning a rate of return that exceeds the rate of time preference, agents remain liquidity constrained in the DM. In equilibrium, the marginal value of the liquidity services of money is negative, $\Delta^{b\prime}(a) < 0$, but this is exactly offset by the negative user cost of holding money, $(\rho - i_m)/(1 + i_m) < 0$.

When $i_m \in (\hat{\imath}, \bar{\imath})$, agents hold sufficient real balances to finance y^* and money serves as a store of value at the margin. In this case, $c < c^*$ solves (22), where the left side represents the marginal surplus from an additional unit of consumption, while the right side captures the opportunity cost in terms of foregone interest relative to the rate of time preference.

Finally, when $i_m = \hat{\imath} > \rho$, there are two symmetric equilibria. In one equilibrium, agents choose $a_1^* < p^*$, and therefore are liquidity-constrained in the DM. In the other equilibrium agents choose $a_2^* > p^*$ and $c < c^*$, and therefore face no liquidity constraint in the DM but they consume less when young.

4.2 The modified Friedman rule

We now turn to the optimal monetary policy where social welfare is measured as the sum of the utilities of all agents within a period,

$$\mathcal{W} = \alpha\sigma\left[u(y) - y\right] + U(c) - c. \tag{23}$$

The first term on the right side corresponds to the trade surpluses in the DM. The second term is the CM surplus generated by the consumption of the young. The consumption of the old generates no surplus. Feasibility requires that $c \leq \bar{h}_1 + \bar{h}_2$. The first-best allocation is $y = y^*$ and $c = c^*$.[19] The next proposition characterizes the monetary policy that maximizes social welfare.

Proposition 5 ***(The modified Friedman rule.)*** *Under Assumption 3:*

1. *Among feasible policies, the optimal interest on currency is $i_m = \hat{\imath} > \rho$, where $\hat{\imath}$ decreases with $\bar{h}_1$ and increases with $\alpha\sigma$. It fails to implement the first best.*

[18] The result that $R > 1 + \rho$ is feasible is at odds with the common view in the literature that deflation above the Friedman rule is not feasible. The logic commonly used in the literature is that if $R > 1 + \rho$, the agent's problem has no solution, i.e., the agent could accumulate unbounded asset holdings and enjoy unbounded utility. In our model, this argument fails because agents have limited endowment, $h \leq \bar{h}_1$.

[19] The social welfare function in (23) is analogous to the one used in LW, which makes it easier to compare policy recommendations. In the online Appendix B4, we show that $\mathcal{W}$ approaches the Golden Rule criterion when β tends to one.

2. *Consider a sequence of economies in which $\bar{h}_1 \nearrow +\infty$ and $\bar{h}_2$ increases as needed so that Assumption 3 continues to hold. The optimal monetary policy approaches the Friedman rule from above, $\hat{\imath} \searrow \rho$, and it implements an allocation that converges to the first best.*

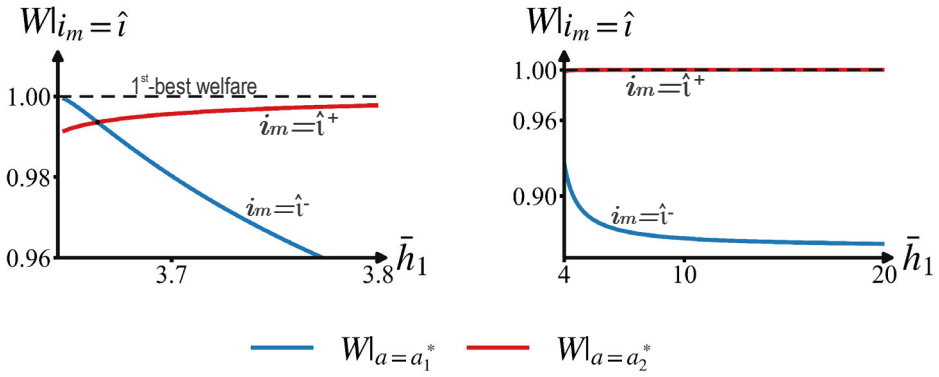


Figure 7: Welfare under the optimal policy, $i_m = \hat{\imath}$, normalized as one minus the welfare loss divided by the absolute value of first-best welfare at the two symmetric equilibria, $a \in \{a_1^*, a_2^*\}$. Functional forms and parameter values are $U(c) = 2.4\log(c)$, $u(y) = y^{0.55}/0.55$, $\rho = 0.05$, $\theta = 0.55$, and $\alpha\sigma = 0.28$. $\bar{h}_2$ satisfies Assumption 3; it does not affect welfare. Left panel: Intermediate values for $\bar{h}_1$ to illustrate how the ranking of equilibria changes. Right panel: Large values for $\bar{h}_1$ to illustrate that the equilibrium with $a = a_2^*$ approaches the first best as $\bar{h}_1$ tends to $+\infty$.

The optimal monetary policy corresponds to an interest rate on currency that exceeds the rate of time preference.[20] It balances the gain from relaxing inefficiently tight liquidity constraints in the DM against the distortion in the intertemporal allocation of CM consumption. At the optimal policy, agents are indifferent between $a_1^* < p^*$ and $a_2^* > R(\bar{h}_1 - c^*)$, giving rise to two symmetric equilibria.

Figure 7 compares welfare across these equilibria as arbitrage capacity, $\bar{h}_1$, varies. When $\bar{h}_1$ is relatively small, the equilibrium in which agents remain liquidity constrained, corresponding to $a = a_1^*$, yields higher welfare. As $\bar{h}_1$ increases, however, the equilibrium with slack liquidity constraints in the DM, corresponding to $a = a_2^*$, becomes preferable. Moreover, welfare levels at the two equilibria diverge as $\bar{h}_1$ grows large. Along the a_1^* branch, asset holdings converge to $\tilde{a} < p^*$ as shown above, so DM output moves farther below y^* and welfare moves farther from the first best. Along the a_2^* branch, DM output remains equal to y^* while CM consumption increases toward c^*, so welfare approaches the first best.

[20] Zhu (2008) reports a similar result in a model without discounting, showing that a contracting money supply can be optimal when buyers lack full bargaining power.

The second part of Proposition 5 considers a sequence of economies in which young agents have an unbounded capacity to purchase assets, $\bar{h}_1 \to +\infty$, while $\bar{h}_2$ increases as needed so that Assumption 3 continues to hold. The optimal monetary policy converges to the Friedman rule. The reason is that when $\bar{h}_1 = +\infty$, agents can make unbounded gains when $i_m > \rho$. The equilibrium that maximizes welfare when $\bar{h}_1$ is large corresponds to $a^* = a_2^*$, as illustrated in the right panel of Figure 7, and the allocation is such that $y = y^*$ and $c < c^*$. At the limit when $\bar{h}_1 \to +\infty$, c approaches c^*. Thus, the optimal policy, which approaches the Friedman rule, implements the first best asymptotically.

In summary, the same environment that generates negative liquidity premia over part of the parameter space overturns a classical policy prescription regarding the optimal rate of return of liquid assets, the Friedman rule. The rate of return of money should be larger than the rate of time preference to compensate for the negative nonpecuniary return of money when liquidity is sufficiently abundant. This result showcases the importance of being explicit about the frictions that generate a role for liquidity.

5 The liquidity structure of asset returns

We now extend the model to multiple assets that differ in their degree of liquidity. This extension serves two complementary purposes. First, it shows that the mechanism generating negative liquidity premia survives when agents can substitute toward less liquid stores of value. Second, it delivers a cross-sectional notion of liquidity premia that corresponds more closely to the empirical measures used in the literature.

We index assets by $j \in \mathbb{J}$. The exogenous supply of asset j is $A^j > 0$. To simplify the exposition, we assume that assets are short-lived: assets created in the CM stage of period $t-1$ mature in the CM stage of period t and yield one unit of the consumption good. Old agents receive the newly created assets as lump-sum endowments. The price of one unit of asset j is denoted by q^j and its gross real rate of return is $R^j = 1/q^j$. We partition the DM meetings into $|\mathbb{J}|$ types. In type-j meetings, only type-j assets can be used as means of payment.[21] The probability of a type-j meeting is α_j and $\alpha = \sum_j \alpha_j$.

[21] This assumption is related to the preferred habitat theory used in finance to explain the yield curve. It posits that investors exhibit preferences for specific maturities or types of bonds due to hedging needs or other considerations. For a formalization of agents' preferences for certain asset classes in a New Monetarist model similar to ours, see, e.g., Jacquet (2021). Online Appendix B7 considers nested acceptability, whereby a more liquid asset can be used wherever a less liquid asset can be used, and shows that our cross-sectional results are robust to this alternative formulation.

Lemma 2 *An agent's optimal portfolio is an element of*

$$\mathbf{A}^*(\mathbf{q}) = \arg\max_{\mathbf{a}\in\mathbb{R}_+^{|\mathbb{J}|}} \left\{\Gamma(\mathbf{a}) : \mathbf{a}.\mathbf{q} \leq \bar{h}_1\right\}, \tag{24}$$

where $\mathbf{q} \equiv (q^j)^\top$ *and*

$$\Gamma(\mathbf{a}) \equiv \max_{c\leq \bar{h}_1 - \mathbf{a}.\mathbf{q}} \left\{ U(c) - c - \mathbf{a}.(\mathbf{q} - \boldsymbol{\beta}) + \beta \sum_{j\in\mathbb{J}} \alpha_j \sigma \Delta^b(a^j) \right\}, \tag{25}$$

for all $\mathbf{a} \in \mathbb{R}_+^{|\mathbb{J}|}$ *such that* $\mathbf{a}.\mathbf{q} \leq \bar{h}_1$*, and where* $\boldsymbol{\beta} \equiv (\beta, ..., \beta)^\top$*.*

The agent chooses her portfolio, $\mathbf{a} \equiv (a^j)$, and her consumption, c, to maximize her CM utility subject to a budget constraint. Market clearing requires that

$$\mathbf{A} \in \text{A}^d(\mathbf{q}) \equiv \text{conv}[\mathbf{A}^*(\mathbf{q})], \tag{26}$$

where $\mathbf{A} \equiv (A^j)$. The left side is the asset supply where we use that assets mature in the CM by paying off one unit of numeraire. The aggregate asset demand is the convex hull of the set of all optimal portfolios. An equilibrium consists of a price vector $\mathbf{q}$ and a probability distribution supported on $\mathbf{A}^*(\mathbf{q})$ whose mean is $\mathbf{A}$. The following proposition establishes existence for any finite vector of positive asset supplies.

Proposition 6 ***(Existence of equilibrium with multiple assets.)*** *For every finite vector of positive asset supplies, an equilibrium of the multiple-asset economy exists.*

5.1 Liquid and illiquid assets

We start with two assets, a liquid asset, $j = 1$, and an illiquid one, $j = 0$, with $\alpha_0 = 0 < \alpha_1$. An illiquid asset is an asset that cannot serve as a medium of exchange in the DM. The liquidity premium of asset 1 is defined as the difference in pecuniary return between asset 1 and asset 0, a difference entirely attributable to $\alpha_1 > 0$. Hence,

$$\mathcal{L}^1 \equiv R^0 - R^1 = \frac{q^1 - q^0}{q^0 q^1}.$$

We show below that negative liquidity premia persist even when agents can substitute toward illiquid assets.

Proposition 7 ***(Equilibria with liquid and illiquid assets.)*** *Assume*

$$\bar{h}_1 > c^* + \beta A^0 + \beta \max_{a\in(0,p^*]} \left[1 + \alpha_1 \sigma \Delta^{b\prime}(a^-)\right] a. \tag{27}$$

There are two thresholds, $0 < \underline{A}^1 < \bar{A}^1$*, such that the following is true:*

1. *For all $A^1 < \underline{A}^1$, there is a symmetric equilibrium where*

$$q^0 = \beta \tag{28}$$
$$q^1 = \beta\left[1 + \alpha_1 \sigma \Delta^{b\prime}(A^1)\right]. \tag{29}$$

All agents hold a portfolio composed of $A^1 < p^*$ units of liquid assets and A^0 units of illiquid assets. Moreover, $\mathcal{L}^1 > 0$ if $A^1 < \tilde{a}$, and $\mathcal{L}^1 < 0$ if $A^1 \in (\tilde{a}, \underline{A}^1)$.

2. *For all $A^1 \in \left(\underline{A}^1, \bar{A}^1\right)$, there exists an asymmetric equilibrium where*

$$q^1 = \beta\left[1 + \alpha_1 \sigma \Delta^{b\prime}(\underline{A}^1)\right] < q^0 = \beta. \tag{30}$$

A fraction ω of agents hold a portfolio composed of $\underline{A}^1 < p^*$ units of liquid assets and A^0/ω units of illiquid assets while the remaining fraction $1-\omega$ hold $\breve{a}^1 > p^*$ units of the liquid asset, where

$$\breve{a}^1 = \arg \max_{a^1 \geq p^*, c \geq 0} \left\{U(c) - c - a^1(q^1 - \beta) : c + a^1 q^1 \leq \bar{h}_1\right\}, \tag{31}$$

and no illiquid asset. The fraction of liquidity-constrained agents is

$$\omega = \frac{\breve{a}^1 - A^1}{\breve{a}^1 - \underline{A}^1}. \tag{32}$$

3. *For all $A^1 > \bar{A}^1$, any equilibrium features $q^1 < q^0 < \beta$ and hence $\mathcal{L}^1 < 0$. Moreover, there is a positive fraction of agents for whom the liquidity constraint binds in the DM.*

Proposition 7 delivers two main results. First, negative liquidity premia are robust to the introduction of illiquid assets. This result is not immediate because agents can substitute illiquid for liquid assets once liquidity premia become negative. One might therefore conjecture that no agent would hold more than $\tilde{a}$ units of the liquid asset, preventing liquidity premia from turning negative. Proposition 7 shows that this intuition is incorrect. Since portfolio choices and asset prices are jointly determined, market clearing requires some agents to hold more than $\tilde{a}$ units of the liquid asset whenever A^1 is larger than $\tilde{a}$, and the equilibrium prices supporting this allocation imply a negative liquidity premium.

A noticeable difference relative to the one-asset economy is that the liquidity premium does not return to zero even for arbitrarily large (finite) values of A^1. Once A^1 has passed a threshold, the liquid asset remains cheaper than the illiquid one, $q^1 < q^0$, irrespective of A^1. The argument is based on the following contradiction. Suppose $A^1 > p^*$. By market clearing, some agents must hold more than p^* and, therefore, must be unconstrained in the DM. If $q^1 = q^0$, unconstrained agents could increase their DM surplus at no cost by replacing

some liquid assets with illiquid ones while keeping the value of their portfolios unchanged. Hence, $q^1 = q^0$ cannot be part of an equilibrium. This argument has no counterpart in the one-asset economy because there is no illiquid asset with which agents can substitute away from excess liquid holdings.

The second main result of Proposition 7 is the endogenous heterogeneity in asset portfolios. In the one-asset economy, we have established that there exist asymmetric equilibria where agents hold different amounts of liquid assets and that an endogenous fraction of agents is liquidity constrained. With multiple assets, agents not only choose different levels of wealth, but also different portfolio compositions between liquid and illiquid assets. The equilibrium becomes asymmetric when A^1 crosses $\underline{A}^1$. Agents who choose to be liquidity constrained in the DM hold both liquid and illiquid assets. They economize on liquid balances and invest the remainder of their wealth in the illiquid asset. Agents who choose to be unconstrained in the DM only hold liquid assets. The DM trade surplus of unconstrained agents is lower than that of constrained agents. However, because the liquid asset is cheaper, the unconstrained agents make up for their loss by holding a large quantity of that asset.

5.2 A quantitative exploration

We now calibrate the model to gauge its ability to generate realistic liquidity premia. We take as our leading example the market for long-maturity Treasuries and use the series of supplies and convenience yields constructed by Jiang et al. (2025), covering January 1996 to January 2024. Long-maturity Treasuries are defined as securities with 7 to 20 years of remaining maturity. Treasury supply is measured as the market value of securities outstanding in this maturity range, expressed as a percentage of GDP. We relate this measure of supply to the 10-year convenience yield, defined as the spread between the 10-year interest-rate swap rate and the 10-year Treasury yield. We adopt a minimalist calibration strategy based on two local moments from Jiang et al. (2025) and conduct several robustness checks.

We take the period to be a year and set $\beta = 0.96$. Although assets in the model mature after one period, maturity is not essential for our mechanism, which depends on the market value of liquid assets available to facilitate transactions rather than the timing of their payoff. We return to asset maturity in a later section. We interpret one unit of A^1 as one percentage point of GDP, treating aggregate GDP as an external scale variable. Because each asset pays one unit of the CM good when it matures, A^1 also measures the aggregate value of liquid assets available for DM trade. The role of the illiquid asset is to price liquidity premia. We therefore set its supply to a small value, $A^0 = 0.01$, so that it has a negligible effect on

agents' labor-supply constraints.[22]

We consider a symmetric benchmark where agents have equal bargaining power in pairwise trades, $\theta = 0.5$. We will later recalibrate the model for asymmetric bargaining powers, $\theta = 0.3$ and $\theta = 0.75$. The utility function in the CM when young is $U(c) = \log(c)$ and the utility function in the DM is $u(y) = By^{1-\eta}$ with $\eta \in (0, 1)$. In the context of Treasury markets, y can be interpreted as the size of a transaction that requires Treasuries for financing or settlement. The total gains from trade associated with the transaction are $u(y) - y$. This specification is sufficiently agnostic to encompass a variety of Treasury-financed or Treasury-facilitated transactions.[23] Evidence from the repo market supports the assumption of diminishing marginal returns from Treasury holdings (e.g., D'Amico, Fan, and Kitsul, 2018). However, it does not directly identify the curvature parameter η. Thus, we consider three distinct values, $\eta \in \{0.5, 0.7, 0.9\}$, beginning with $\eta = 0.7$ as our benchmark.

We calibrate the two remaining parameters, B and $\alpha\sigma$, to two local moments taken from Jiang et al. (2025), rather than fitting the entire empirical relationship between Treasury supply and convenience yields. The first target is the mean ten-year convenience yield in their benchmark sample, $\mathcal{L}^1 = 27.4$ basis points, which we associate with the corresponding mean long-term Treasury supply, $A^{1*} = 4.79$ percent of GDP. The second target is the local semi-elasticity of the convenience yield with respect to supply obtained from an IV regression, $d\mathcal{L}^1/d\log A^1$, which Jiang et al. (2025) estimate to be equal to -1.316. We interpret this estimate as a steady-state derivative in the model and therefore impose as our two key moments,

$$\mathcal{L}^1(A^{1*}) = 27.4 \text{ bp} \quad \text{and} \quad A^{1*}\mathcal{L}^{1\prime}(A^{1*}) = -131.6 \text{ bp},$$

where convenience yields are measured in basis points. The sample mean of the convenience yield anchors its average level, while the IV estimate disciplines its response to exogenous variation in Treasury supply. This lets us assess separately whether the calibrated model can account for the broader empirical relationship.

[22] A 100-fold increase in A^0 from 0.01 to 1 leaves the calibration unchanged and the out-of-sample predictions are also remarkably stable.

[23] One interpretation is that the buyer is an institutional investor, such as a hedge fund or dealer, with access to a trading or investment opportunity of scale y, yielding payoff $u(y)$. The opportunity may take the form of an arbitrage trade, a hedging position, or another investment. Undertaking the position requires financing, for which Treasury securities can serve as collateral. The financing may be provided by a dealer, bank, or money-market fund, while the underlying transaction takes place with a separate counterparty. The model consolidates these counterparties into a single seller. Thus, Treasury holdings determine the investor's capacity to finance the transaction, while $u(y)$ captures the payoff from the underlying economic opportunity. The gains from trade do not originate from Treasury holdings themselves: Treasuries facilitate the financing needed to realize those gains. This interpretation is consistent with the use of Treasuries as collateral in repo markets to finance market making, hedging, arbitrage, and other trading activities; see, e.g., D'Amico, Fan, and Kitsul (2018).

Table 1: Benchmark calibration

Parameter	Value	Calibration strategy
β	0.96	Annual discount factor
θ	0.50	Symmetric Nash bargaining
η	0.70	Benchmark curvature; robustness considered
A^0	0.01	Small illiquid-asset supply
B	13.07	Convenience-yield level and semi-elasticity
$\alpha\sigma$	0.0227	Convenience-yield level and semi-elasticity
$\bar{h}_1$	25	Arbitrage capacity; dealer balance-sheet evidence

Finally, we interpret $\bar{h}_1 - c^*$ as agents' arbitrage capacity—the resources beyond consumption needs that agents can deploy to absorb Treasury securities in the CM. Empirically, we associate this capacity with primary dealers' balance-sheet capacity, since dealer banks are important absorbers of Treasury supply in practice. Their willingness to warehouse securities is constrained in part by regulatory leverage limits. As an order-of-magnitude benchmark, we use the estimate in Cochran et al. (2024) that, as of June 2024, bank holding companies associated with primary dealers had approximately \$7.3 trillion of unused balance-sheet capacity, or about 25 percent of annual GDP, and set $\bar{h}_1 = 25$. (We ignore $c^* = 1$ in this calculation as it is small relative to $\bar{h}_1$.) We view this estimate as a loose upper-bound proxy since it does not measure the capacity specifically available for Treasury intermediation. We therefore consider alternative values of $\bar{h}_1$ in our sensitivity analysis. At the calibration point, the labor-supply constraint of young agents is slack, so $\bar{h}_1$ does not affect the two moments used to determine B and $\alpha\sigma$. From the two targets above we obtain $B = 13.0727$ and $\alpha\sigma = 0.02273$.[24]

The empirical moments used in the calibration are estimated with some uncertainty. To assess its implications for our results, we follow the bootstrap procedure of Jiang et al. (2025) and report the resulting 90 percent confidence band in Figure 8. Online Appendix B8 provides details on the bootstrap procedure and the construction of the confidence band.

As shown in the top left panel of Figure 8, despite targeting only two local moments, the model captures the overall relationship between Treasury supply and convenience yields well over the observed range of Treasury supplies. The substantial variation in convenience yields around the model prediction likely reflects factors other than Treasury supply—most notably monetary policy (Nagel, 2016)—that lie outside the scope of the model. Moreover, the calibration does not target the point at which the liquidity premium changes sign. The

[24] Cochran et al. (2024) measure unused balance-sheet capacity as the additional leverage exposure that these banking organizations could accommodate, holding capital fixed, before reaching their regulatory minimum leverage ratio. They emphasize that, under normal market conditions, minimum Supplementary Leverage Ratio (SLR) requirements were not expected to constrain dealer intermediation.

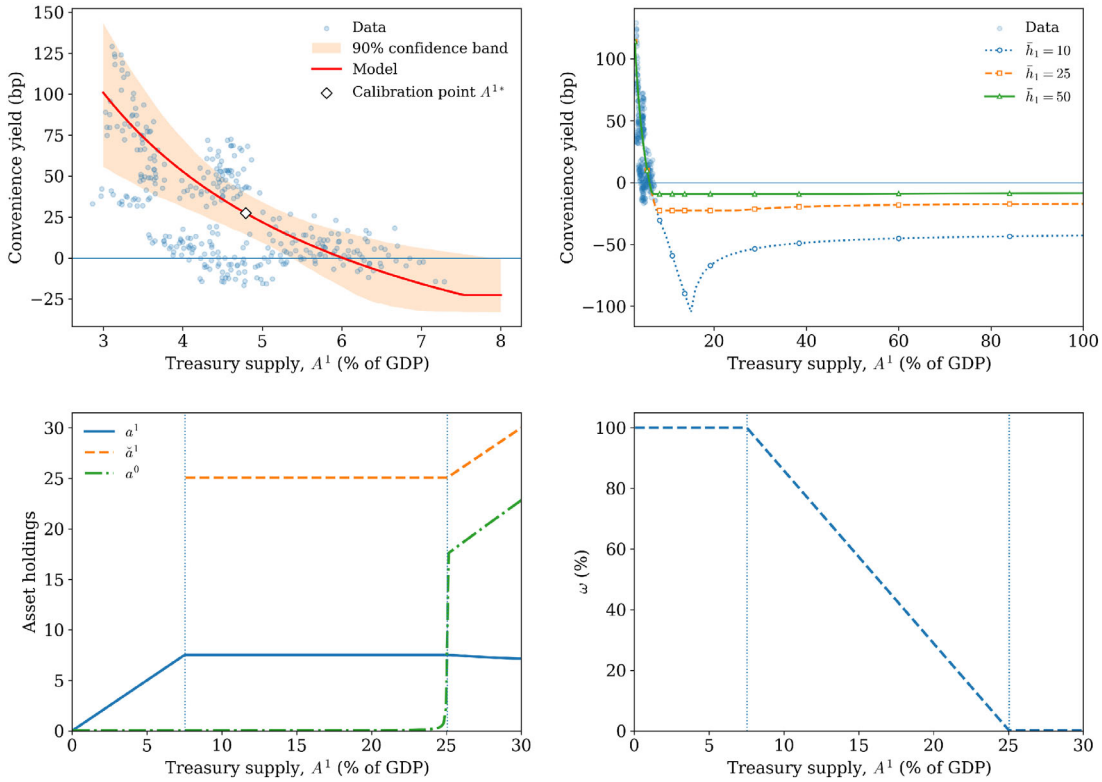


Figure 8: Top left: Model fit within the observed range of Treasury supplies. The blue dots correspond to the raw data. The diamond denotes the calibration point constructed from the benchmark long-term Treasury sample in Jiang et al. (2025). The red curve is the model prediction. Top right: Out-of-sample model predictions at large asset supplies. Bottom left: Individual asset portfolios. Liquidity-constrained agents hold (a^0, a^1) while unconstrained agents hold $(0, \breve{a}^1)$. Bottom right: Fraction of liquidity-constrained agents.

calibrated curve crosses zero at a supply of approximately 6.0 percent of GDP, with a 90 percent bootstrap interval for the zero-crossing threshold of approximately 5.4% to 7.9%.

To examine the model's out-of-sample predictions at large asset supplies, we consider $\bar{h}_1 \in \{10, 25, 50\}$. The top right panel of Figure 8 shows that negative convenience yields persist at very large asset supplies for all three values of arbitrage capacity. Greater arbitrage capacity attenuates the negative premium but does not eliminate it.

The bottom panels of Figure 8 illustrate the endogenous portfolio heterogeneity that emerges as the supply of the liquid asset increases. At low supplies, all agents choose the same portfolio and are liquidity constrained. Once A^1 reaches a threshold, the equilibrium becomes asymmetric: a shrinking fraction of liquidity-constrained agents hold both liquid and illiquid assets, while a growing fraction of unconstrained agents emerges and specializes in the liquid asset. At sufficiently large supplies, the labor-supply constraint of the young

binds, but the fraction of liquidity-constrained agents remains positive, although it becomes very small.

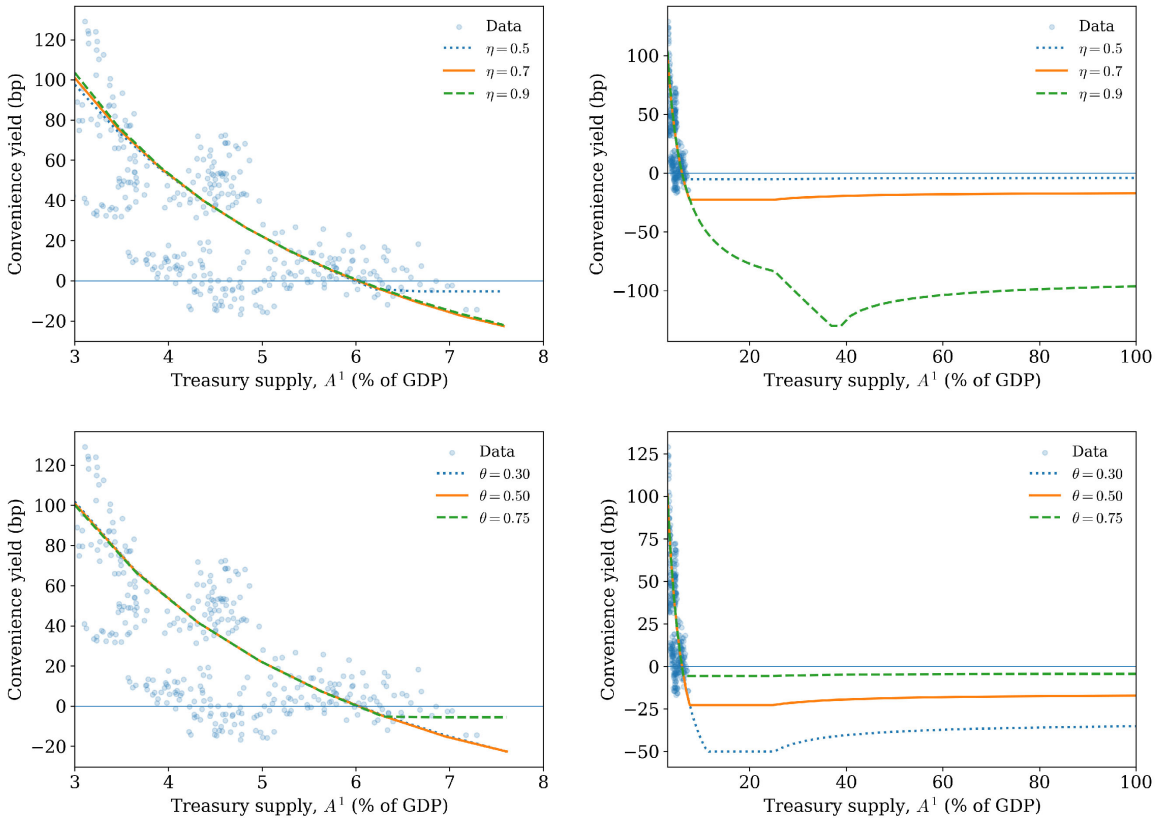


Figure 9: Model is recalibrated for different values of η and θ. Top: $\eta \in \{0.5, 0.7, 0.9\}$. Bottom: $\theta \in \{0.3, 0.5, 0.75\}$.

Finally, we recalibrate our model for different degrees of curvature of DM utility, $\eta \in \{0.5, 0.7, 0.9\}$, and different values of the bargaining power, $\theta \in \{0.3, 0.5, 0.75\}$. The general fit of the model and the out-of-sample predictions are shown in Figure 9. As shown by the left panels, the model continues to capture the overall relationship between Treasury supply and convenience yields for all values of η and θ. The key difference concerns the magnitude of negative liquidity premia out of sample. For higher values of η and lower values of θ, liquidity premia can become substantially more negative over an intermediate range of A^1. A richer quantitative analysis would incorporate Treasuries of different maturities as well as monetary policy. But our first pass suggests that the model can account for the sign and magnitude of liquidity premia.

5.3 Liquidity and asset returns

We conclude our analysis of multiple assets by addressing two questions. First, how do differences in assets' liquidity characteristics, captured by α_j, interact with asset supplies to shape the distribution of liquidity premia across assets? Second, can positive and negative liquidity premia coexist? The first question is motivated by the view that convenience yields reflect different dimensions of liquidity, including an asset's ease of trade and its moneyness. The second is motivated by the evidence in Fleckenstein and Longstaff (2024) and Jiang et al. (2025), illustrated in Figure 1, showing that different Treasury securities can exhibit liquidity premia of opposite signs.

The following proposition answers these questions by characterizing equilibria of an economy that has three assets, $\mathbb{J} = \{0, 1, 2\}$, with $\alpha_0 = 0 < \alpha_1 < \alpha_2$. Thus, asset 0 is illiquid and asset 2 is the most liquid one. We focus on symmetric equilibria where the labor supply constraint, $h \leq \bar{h}_1$, is slack. Hence, asset prices are given by the first-order (necessary but not sufficient) conditions of (25),

$$q^j = \beta \left[1 + \alpha_j \sigma \Delta^{b\prime}(A^j)\right] \quad \forall j \in \{1, 2\}. \tag{33}$$

Asset prices in (33) reflect the interaction between an asset's acceptability and its supply.

Proposition 8 ***(Liquidity-based yield curve.)*** *Suppose $\bar{h}_1$ is sufficiently large.*

1. ***Conventional liquidity-based yield curve.*** *There exists an equilibrium featuring*

$$q^2 > q^1 > q^0 = \beta,$$

i.e., liquidity premia are positive and increase with acceptability, if

$$\alpha_2 \Delta^{b\prime}(A^2) > \alpha_1 \Delta^{b\prime}(A^1) > 0. \tag{34}$$

2. ***Mixed liquidity premia.*** *For $j, j' \in \{1, 2\}$, $j \neq j'$, there exists $\breve{a}(\bar{h}_1, A^j) > \tilde{a}$ such that, whenever $A^j < \tilde{a}$ and $A^{j'} \in (\tilde{a}, \breve{a})$, there exists an equilibrium featuring*

$$q^j > q^0 = \beta > q^{j'},$$

i.e., the liquidity premium of asset j is positive while that of asset j' is negative.

3. ***Inverted liquidity-based yield curve.*** *There exists $\breve{a}(\bar{h}_1) > \tilde{a}$ such that, whenever $(A^1, A^2) \in (\tilde{a}, \breve{a})^2$ satisfies*

$$\alpha_2 \Delta^{b\prime}(A^2) < \alpha_1 \Delta^{b\prime}(A^1) < 0, \tag{35}$$

there exists an equilibrium featuring

$$q^0 = \beta > q^1 > q^2,$$

i.e., liquidity premia are negative and inversely related to acceptability.

The first part of Proposition 8 establishes the existence of equilibria featuring a conventional liquidity-based yield curve where positive liquidity premia are ordered by assets' acceptabilities. Such an equilibrium exists when the supplies of both liquid assets are sufficiently low, and in particular when the supply of the more liquid asset is not too large relative to that of the less liquid asset. The second part of Proposition 8 shows that positive and negative liquidity premia can coexist in equilibrium. For this, the supply of the asset j' is above the threshold $\tilde{a}$ that turns positive premia into negative ones while the supply of j is below that threshold. The supply $A^{j'}$ is required not to be too large so that the constraint $h \leq \bar{h}_1$ stays slack. The last part of Proposition 8 generates a full inversion of the liquidity-based yield curve. When the supplies of all liquid assets become sufficiently large, the marginal value of liquidity services turns negative for every liquid asset.

5.4 Maturity-segmented markets

An important observation from Jiang et al. (2025) is that liquidity premia tend to decrease with maturity. Our framework admits a simple interpretation as a maturity-segmented market with preferred habitats. Suppose there are different securities indexed by j that differ according to their maturity. Only a fraction n_j of investors participates in the market for securities of maturity j, as in preferred-habitat theories. The relevant supply of maturity-j securities is then their effective supply per participating investor, A^j/n_j.

Maturity-j securities pay one unit of numeraire good each period and mature with probability $\delta_j \in (0, 1]$ at the end of each period. Accordingly, $1/\delta_j$ provides a measure of the asset's expected maturity. To maintain a constant aggregate supply, the old receive a flow $\delta_j A^j$ of newly issued assets as lump-sum endowments in the CM of each period, after existing assets pay out and maturing assets are retired, and before trade with the young. The gross real rate of return of maturity-j securities is

$$R^j = \frac{1 + (1 - \delta_j)q^j}{q^j} \Leftrightarrow q^j = \frac{1}{R^j - (1 - \delta_j)}.$$

It follows that the market capitalization of the maturity-j securities in the DM is

$$\left[(1 - \delta_j)q^j + 1\right] A^j = \frac{R^j A^j}{R^j - (1 - \delta_j)}. \tag{36}$$

Consider equilibria where the constraint $h \leq \bar{h}_1$ is slack for all investors. By the same reasoning as above and using that market capitalization is given by (36), the rate of return of maturity-j securities solves

$$\frac{1-\beta R^j}{\beta R^j} = \alpha_j \sigma \Delta^{b\prime}\left(\frac{R^j}{R^j-(1-\delta_j)}\frac{A^j}{n_j}\right). \tag{37}$$

Equation (37) determines R^j as a function of asset supply, A^j, habitat size, n_j, and intrinsic liquidity, α_j.

Proposition 9 (Maturity-segmented markets) *The threshold asset supply at which the liquidity premium changes sign is*

$$A_0^j = n_j\left(\frac{\rho+\delta_j}{1+\rho}\right)\tilde{a}.$$

It is increasing in n_j and δ_j.

The result follows immediately from (37). Holding outstanding supply fixed, a smaller investor habitat (lower n_j) raises the effective supply per participating investor and therefore pushes the liquidity premium toward the negative region. A longer maturity (lower δ_j) has the same effect: for a given number of outstanding securities, it raises their market capitalization and hence the effective supply of liquidity per participating investor, pushing liquidity premia toward negative values.

6 Conclusion

We showed that the canonical New Monetarist model of asset liquidity naturally generates both positive and negative liquidity premia. Negative liquidity premia emerge when the supply of liquid assets rises above an endogenous threshold. We discussed in detail the role of each assumption in generating this result, including the trading mechanism, the information structure, and agents' lack of commitment. We calibrated our model to the market for long-maturity U.S. Treasuries and showed that the model can account for the sign and magnitude of liquidity premia observed in the data. From a methodological standpoint, we showed the usefulness of an overlapping-generations version of the New Monetarist model for characterizing equilibria over the entire parameter space, thereby uncovering novel predictions of the model. Beyond explaining the origin of negative liquidity premia, these predictions include the endogenous sorting of ex ante identical agents into liquidity-constrained and unconstrained positions and an optimal rate of return on money above the Friedman rule.

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MAIN APPENDIX

A1. Proofs of lemmas and propositions

Lemma 3 *Under Assumptions 1a and 2,*

$$g_1(a;R) = U(c^*) - c^* + \beta\left\{-\left(\frac{1-\beta R}{\beta R}\right)a + \alpha\sigma\Delta^b(a)\right\} \quad \forall a \in [0,p^*]. \tag{38}$$

The correspondence $\text{A}_1^*(R) \equiv \arg\max_{a\in[0,p^*]} g_1(a;R)$ *is upper hemicontinuous and any* $a_1^* \in \text{A}_1^*$ *is a solution to*

$$-\left(\frac{1-\beta R}{\beta R}\right) + \alpha\sigma\Delta^{b\prime}(a_1^*) \geq 0, \quad \text{“}=\text{”} \text{ if } a_1^* < p^*, \tag{39}$$

where $\Delta^{b\prime}(a_1^*) = \Delta^{b\prime}(p^{*-})$ *if* $a_1^* = p^*$. *Any selection from* $\text{A}_1^*(R)$ *is non-decreasing in* R, *and strictly increasing when solutions are interior. Moreover,* $a_1^* \in (0,\tilde{a})$ *for all* $R < \beta^{-1}$; $a_1^* = \tilde{a}$ *if* $R = \beta^{-1}$; *and* $a_1^* \in (\tilde{a}, p^*]$ *for all* $R > \beta^{-1}$.

Proof of Lemma 3. Expression for the function g_1. Suppose the constraint, $c \leq \bar{h}_1 - a/R$, in (7) is slack. The first-order condition with respect to c gives $U'(c) = 1$, i.e., $c = c^*$. Substituting $c = c^*$ into (7), one obtains (38). Given $c = c^*$, the constraint is slack if $\bar{h}_1 \geq c^* + a/R$. In any stationary equilibrium, $R \geq 1$. Thus, for the inequality $c^* \leq \bar{h}_1 - a/R$ to hold for all $a \in [0,p^*]$, it is sufficient to assume that $\bar{h}_1 \geq c^* + p^*$, which corresponds to Assumption 2.

The optimal-choice correspondence. By Assumption 1a, the function g_1 is continuous. Thus, by the Maximum Theorem, $\text{A}_1^*(R) \equiv \arg\max_{a\in[0,p^*]} g_1(a;R)$ is non-empty and upper hemicontinuous. Since g_1 is differentiable on $(0,p^*)$, from the first-order condition, any optimal choice, a_1^*, solves (39). Note first that $a_1^* \neq p^*$ when $R \leq \beta^{-1}$. Indeed, $\Delta^{b\prime}(p^{*-}) < 0$ and hence the inequality in (39) cannot hold. Also, by Assumption 1a, Δ^b is strictly unimodal, $\Delta^{b\prime}(a) > 0$ for all $a \in (0,\tilde{a})$ with $\Delta^{b\prime}(0) = +\infty$ and $\Delta^{b\prime}(\tilde{a}) = 0$. Thus, if $R = \beta^{-1}$, (39) implies $\Delta^{b\prime}(a_1^*) = 0$, i.e., $a_1^* = \tilde{a}$. For all $R < \beta^{-1}$, $(1-\beta R)/\beta R > 0$ and hence, from (39) at equality, any $a_1^* \in \text{A}_1^*$ is such that $a_1^* \in (0,\tilde{a})$. By the same logic, if $R > \beta^{-1}$, $\Delta^{b\prime}(a_1^*) < 0$, and hence $a_1^* \in (\tilde{a}, p^*]$.

Monotonicity of the optimal choice. We now establish the monotonicity of $\text{A}_1^*(R)$. Consider $R' > R$, $a_1^* \in \text{A}_1^*(R)$ and $a_1^{*\prime} \in \text{A}_1^*(R')$. By revealed preferences:

$$\left(\frac{1-\beta R'}{\beta R'}\right)(a_1^{*\prime} - a_1^*) \leq \alpha\sigma\left[\Delta^b(a_1^{*\prime}) - \Delta^b(a_1^*)\right] \leq \left(\frac{1-\beta R}{\beta R}\right)(a_1^{*\prime} - a_1^*).$$

Using that $(1-\beta R')/\beta R' < (1-\beta R)/\beta R$, it follows immediately that $a_1^{*\prime} \geq a_1^*$. Suppose $a_1^{*\prime} < p^*$. From (39) at equality,

$$-\left(\frac{1-\beta R}{\beta R}\right) + \alpha\sigma\Delta^{b\prime}(a_1^*) = 0 = -\left(\frac{1-\beta R'}{\beta R'}\right) + \alpha\sigma\Delta^{b\prime}(a_1^{*\prime}).$$

Equalizing the left and right sides and rearranging, we obtain

$$\alpha\sigma\left[\Delta^{b\prime}(a_1^*) - \Delta^{b\prime}(a_1^{*\prime})\right] = \left(\frac{1-\beta R}{\beta R}\right) - \left(\frac{1-\beta R'}{\beta R'}\right) > 0.$$

This rules out $a_1^{*\prime} = a_1^*$. Thus, $a_1^{*\prime} > a_1^*$, i.e., a_1^* is increasing in R. ■

Lemma 4 *Under Assumption 2,*

$$g_2(a;R) = U(c^*) - c^* - \beta\left(\frac{1-\beta R}{\beta R}\right)a + \beta\alpha\sigma\Delta^b(p^*) \quad \text{if } a \in \left[p^*, R\left(\bar{h}_1 - c^*\right)\right] \tag{40}$$

$$= U\left(\bar{h}_1 - \frac{a}{R}\right) - \bar{h}_1 + \beta a + \beta\alpha\sigma\Delta^b(p^*) \qquad \text{if } a \in \left[R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right]. \tag{41}$$

Moreover,

$$\text{A}_2^*(R) \equiv \arg\max_{a\in[p^*,R\bar{h}_1]} g_2(a;R) = \begin{cases} \{p^*\} \\ \left[p^*, R\left(\bar{h}_1 - c^*\right)\right] \\ \left\{R\left[\bar{h}_1 - U'^{-1}(\beta R)\right]\right\} \end{cases} \quad \text{if } R \begin{cases} < \\ = \ \beta^{-1} \\ > \end{cases} \tag{42}$$

Proof of Lemma 4. The function g_2. Using that $\Delta^b(a) = \Delta^b(p^*)$ for all $a \geq p^*$, (7) can be rewritten as

$$g_2(a) = \max_{c\leq\bar{h}_1 - a/R}\left\{U(c) - c - \left(\frac{1-\beta R}{R}\right)a + \beta\alpha\sigma\Delta^b(p^*)\right\}.$$

If the constraint on c does not bind, $c = c^*$ and we obtain (40); otherwise, we substitute $c = \bar{h}_1 - a/R$ into the objective to obtain (41). The constraint $c \leq \bar{h}_1 - a/R$ is not binding if $c^* \leq \bar{h}_1 - a/R$, i.e., $a \leq R\left(\bar{h}_1 - c^*\right)$. By Assumption 2 and $R \geq 1$, the interval $\left[p^*, R\left(\bar{h}_1 - c^*\right)\right]$ is nonempty.

The correspondence $\text{A}_2^*(R)$. From (40)-(41),

$$g_2'(a) = \begin{cases} -(1-\beta R)/R & \text{if } a \in \left(p^*, R\left(\bar{h}_1 - c^*\right)\right] \\ -\left[U'\left(\bar{h}_1 - \frac{a}{R}\right) - \beta R\right]/R & \text{if } a \in \left(R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right]. \end{cases}$$

The left and the right derivatives coincide at $a = R\left(\bar{h}_1 - c^*\right)$ since $U'(c^*) = 1$. Thus, g_2 is continuously differentiable. Moreover,

$$g_2''(a) = \begin{cases} 0 & \text{if } a \in \left(p^*, R\left(\bar{h}_1 - c^*\right)\right) \\ U''\left(\bar{h}_1 - \frac{a}{R}\right)/R^2 & \text{if } a \in \left(R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right). \end{cases}$$

The function g_2 is continuous and concave on the interval $\left[p^*, R\bar{h}_1\right]$. Hence, $\mathrm{A}_2^*(R)$ is non-empty and convex-valued. Upper hemicontinuity follows from the Maximum Theorem. The function g_2 is linear on $\left(p^*, R\left(\bar{h}_1 - c^*\right)\right]$ and strictly concave on $\left(R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right]$. Thus, if there is a solution such that $a > R\left(\bar{h}_1 - c^*\right)$, it is unique. Using that $U'(0) = +\infty$, $g_2'(a) \to -\infty$ as $a \to R\bar{h}_1$, so $a_2^* < R\bar{h}_1$. The first-order condition for an optimum is necessary and sufficient and is given by

$$g_2'(a_2^*) \leq 0, \quad \text{“}=\text{”} \quad \text{if } a_2^* > p^*.$$

It follows that any optimal solution, $a_2^* \in \mathrm{A}_2^*(R)$, is such that:

$$a_2^* \begin{cases} = p^*, & \beta R < 1 \\ \in [p^*, R\left(\bar{h}_1 - c^*\right)], & \beta R = 1 \\ = R\left[\bar{h}_1 - U'^{-1}(\beta R)\right], & \beta R > 1. \end{cases}$$

Indeed, if $\beta R < 1$, $g_2'(a) < 0$ for all $a \geq p^*$, in which case $a_2^* = p^*$. If $\beta R = 1$, $g_2'(a) = 0$ for all $a \in \left[p^*, R\left(\bar{h}_1 - c^*\right)\right]$ and $g_2'(a) < 0$ for all $a > R\left(\bar{h}_1 - c^*\right)$. If $\beta R > 1$, $g_2'(a) > 0$ for all $a \in \left[p^*, R\left(\bar{h}_1 - c^*\right)\right]$. Since $g_2(a)$ is strictly concave over the interval $\left(R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right)$, and using the assumption $U'(0) = +\infty$, the unique solution belongs to the interval $\left(R\left(\bar{h}_1 - c^*\right), R\bar{h}_1\right)$. ■

Proof of Proposition 1. The proof is based on Lemmas 3 and 4. We define the value functions $\upsilon_1(R) \equiv \max_{a \in [0, p^*]} g_1(a; R)$ and $\upsilon_2(R) \equiv \max_{a \in [p^*, R\bar{h}_1]} g_2(a; R)$. Then, we have:

$$\max_{a \in \left[0, R\bar{h}_1\right]} \Gamma(a) = \max\left\{\upsilon_1(R), \upsilon_2(R)\right\}, \tag{43}$$

and the set of maximizers is such that

$$\mathrm{A}^*(R) = \begin{cases} \mathrm{A}_1^*(R) \\ \mathrm{A}_2^*(R) \\ \mathrm{A}_1^*(R) \cup \mathrm{A}_2^*(R) \end{cases} \quad \text{if } \upsilon_1(R) \begin{cases} > \\ < \\ = \end{cases} \upsilon_2(R). \tag{44}$$

First, we establish that $\upsilon_1(R)$ and $\upsilon_2(R)$ only intersect once. Second, we establish that any selection from A^* is increasing in R.

Part 1: Single crossing. We now show that the value functions, $\upsilon_1(R)$ and $\upsilon_2(R)$ only cross once at some $R = \hat{R}$. In Lemmas 3 and 4 we have established the joint continuity of $g_1(a; R)$ and $g_2(a; R)$. Since their feasible-set correspondences are continuous and compact-valued, the Maximum Theorem implies that $\upsilon_1(R)$ and $\upsilon_2(R)$ are continuous functions of R. For all $R \leq \beta^{-1}$, $\upsilon_1(R) > \upsilon_2(R)$. To see this, we establish that if $\beta R \leq 1$, it is optimal to choose asset holdings no greater than p^*, i.e., $\mathrm{A}^* = \mathrm{A}_1^*$. From (42), if $\beta R \leq 1$,

$\upsilon_2(R) = g_2(p^*) = g_1(p^*)$. From Lemma 3, any optimal solution satisfies $a_1^* \in (0, \tilde{a}]$ and hence $g_1(p^*) < \upsilon_1(R)$. Thus, $\upsilon_2(R) < \upsilon_1(R)$. Next, we establish that for R large enough, $\upsilon_1(R) < \upsilon_2(R)$. In order to see this, note from (38) that

$$g_1(a) \leq U(c^*) - c^* + \beta \left[a + \alpha\sigma\Delta^b(a)\right] \quad \forall a \in [0, p^*],$$

where we used that the term $-(1-\beta R)/\beta R$ is increasing in R and approaches 1 as $R \to \infty$. Hence, $g_1(a)$ is bounded above by $U(c^*) - c^* + \beta \left[p^* + \alpha\sigma\Delta^b(\tilde{a})\right]$. In contrast, from (40),

$$\upsilon_2(R) \geq g_2(R\left(\bar{h}_1 - c^*\right)) = U(c^*) - c^* - (1-\beta R)\left(\bar{h}_1 - c^*\right) + \beta\alpha\sigma\Delta^b(p^*).$$

Using that $\lim_{R\to+\infty}(\beta R - 1)\left(\bar{h}_1 - c^*\right) = +\infty$, as $R \to +\infty$, the right side goes to $+\infty$. Thus, $\upsilon_1(R)$ is bounded above uniformly in R, whereas

$$\lim_{R\to+\infty} \upsilon_2(R) = +\infty.$$

Hence, $\upsilon_2(R) > \upsilon_1(R)$ for all sufficiently large R. Using this result together with $\upsilon_1(\beta^{-1}) > \upsilon_2(\beta^{-1})$, there exists at least one finite value for $R > \beta^{-1}$, denoted $\hat{R}$, such that $\upsilon_1(R) = \upsilon_2(R)$.

Now we prove that $\hat{R}$ is unique. It has been established above that $\upsilon_1(R) = \upsilon_2(R)$ can only occur when $\beta R > 1$, in which case $a_2^*(R) > p^* \geq a_1^*(R)$, where $a_1^*(R) \in \mathrm{A}_1^*(R)$ and $a_2^*(R) \in \mathrm{A}_2^*(R)$. We rule out $a_1^*(R) = p^*$ for a value of R such that $\upsilon_1(R) = \upsilon_2(R)$. If that were the case, $\upsilon_1(R) = g_1(p^*) = g_2(p^*) = \upsilon_2(R)$. From Lemma 4 and the fact that any optimal solution, a_2^*, is greater than p^* when $\beta R > 1$, $g_2(p^*) < \upsilon_2(R)$. Hence, we have a contradiction. Thus, $a_1^*(R) < p^*$. By the Envelope Theorem:

$$\begin{aligned}
\frac{\partial g_1\left(a_1^*(R), R\right)}{\partial R} &= \frac{a_1^*(R)}{R^2} \quad \text{for any } a_1^*(R) \in \mathrm{A}_1^*(R) \\
\frac{\partial g_2\left(a_2^*(R), R\right)}{\partial R} &= U'\left(\bar{h}_1 - \frac{a_2^*(R)}{R}\right)\frac{a_2^*(R)}{R^2} \quad \text{for any } a_2^*(R) \in \mathrm{A}_2^*(R).
\end{aligned}$$

By Corollary 4 (ii) in Milgrom and Segal (2002), the left (right) derivative of $\upsilon_1(R)$ equals the minimum (maximum) of the partial derivative over $\mathrm{A}_1^*(R)$:

$$\begin{aligned}
\upsilon_1'(R^-) &= \frac{a_1^-(R)}{R^2} \quad \text{where } a_1^-(R) = \min \mathrm{A}_1^*(R) \\
\upsilon_1'(R^+) &= \frac{a_1^+(R)}{R^2} \quad \text{where } a_1^+(R) = \max \mathrm{A}_1^*(R).
\end{aligned}$$

Likewise, the left and right derivatives of $\upsilon_2(R)$ are

$$\begin{aligned}
\upsilon_2'(R^-) &= U'\left(\bar{h}_1 - \frac{a_2^-(R)}{R}\right)\frac{a_2^-(R)}{R^2} \quad \text{where } a_2^-(R) = \min \mathrm{A}_2^*(R) \\
\upsilon_2'(R^+) &= U'\left(\bar{h}_1 - \frac{a_2^+(R)}{R}\right)\frac{a_2^+(R)}{R^2} \quad \text{where } a_2^+(R) = \max \mathrm{A}_2^*(R).
\end{aligned}$$

Using that $U'(c) \geq 1$, which follows from $c = \bar{h}_1 - \frac{a_2^*(R)}{R} \leq c^*$, we have

$$v_1'(R^-) \leq v_1'(R^+) < v_2'(R^-) \leq v_2'(R^+).$$

Thus, at every R such that $v_1(R) = v_2(R)$, both one-sided derivatives of $v_1(R) - v_2(R)$ are strictly negative. Hence, $v_1(R) - v_2(R)$ can cross zero only from above. Therefore, there is at most one crossing, since a second crossing would have to approach zero from below. As a result, there is a unique R, denoted $\hat{R}$, such that $v_1(R) = v_2(R)$. See Figure 10 for an illustration of the argument. For all $R < \hat{R}$, $v_1(R) > v_2(R)$ and $\mathrm{A}^*(R) = \mathrm{A}_1^*(R)$. For all $R > \hat{R}$, $v_1(R) < v_2(R)$ and $\mathrm{A}^*(R) = \mathrm{A}_2^*(R)$.

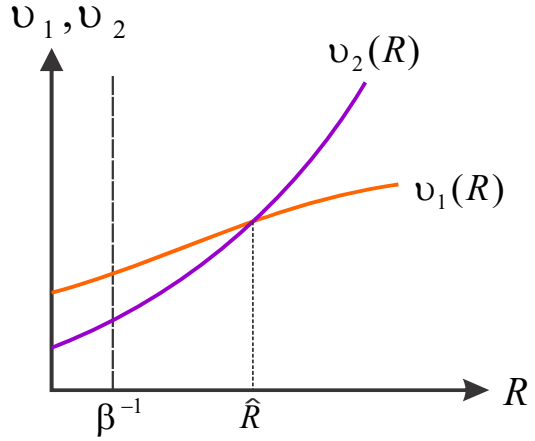


Figure 10: Proof of Proposition 1. The value functions $v_1(R)$ and $v_2(R)$ intersect once at $R = \hat{R}$.

We now establish the comparative statics of $\hat{R}$ using the single-crossing property.

Consider first an increase in $\bar{h}_1$. The value $v_1(R)$ does not depend on $\bar{h}_1$. Since $\hat{R} > \beta^{-1}$, from (42), $a_2^* = R\left[\bar{h}_1 - U'^{-1}(\beta R)\right]$. Substituting this expression into (41) shows that, for a fixed $R > \beta^{-1}$, $v_2(R)$ is strictly increasing in $\bar{h}_1$ since its coefficient on $\bar{h}_1$ is $\beta R - 1 > 0$. Hence, following an increase in $\bar{h}_1$, $v_1(\hat{R}) - v_2(\hat{R}) < 0$ at the original $\hat{R}$. By the single-crossing property, the new crossing must lie to the left of the original one. Thus, $\hat{R}$ is strictly decreasing in $\bar{h}_1$.

Consider next an increase in $\alpha\sigma$. At the original crossing, take any $a_1^* \in \mathrm{A}_1^*(\hat{R})$. Since $\hat{R} > \beta^{-1}$, Lemma 3 implies $a_1^* > \tilde{a}$, and the argument above establishes that $a_1^* < p^*$. Hence, $\Delta^b(a_1^*) > \Delta^b(p^*)$. Following an increase from $\alpha\sigma$ to $(\alpha\sigma)' > \alpha\sigma$, optimality of a_1^* at the original value of $\alpha\sigma$ implies

$$v_1\left(\hat{R}; (\alpha\sigma)'\right) - v_1\left(\hat{R}; \alpha\sigma\right) \geq \beta\left[(\alpha\sigma)' - \alpha\sigma\right]\Delta^b(a_1^*) > \beta\left[(\alpha\sigma)' - \alpha\sigma\right]\Delta^b(p^*),$$

where the last term is exactly the increase in $v_2(\hat{R})$. Thus, $v_1(\hat{R}) - v_2(\hat{R}) > 0$ at the original $\hat{R}$ following the increase in $\alpha\sigma$. By the single-crossing property, the new crossing must lie to the right of the original one. Hence, $\hat{R}$ is strictly increasing in $\alpha\sigma$.

Part 2: Monotonicity of A^*. We established in the proof of Lemma 3 that any selection from A_1^* is increasing in R, and strictly increasing whenever the solution is interior. For $R < \hat{R}$, the relevant solution is indeed interior. If $R > \beta^{-1}$ and $a_1^*(R) = p^*$, then $v_1(R) = g_1(p^*) = g_2(p^*) < v_2(R)$, which contradicts $R < \hat{R}$. (For $R \leq \beta^{-1}$, interiority follows from Lemma 3.) Hence any selection from $\mathrm{A}_1^*(R)$ is strictly increasing for $R < \hat{R}$. From (42), if $\beta R > 1$, a_2^* is strictly increasing in R. From Part 1, for all $R < \hat{R}$, $\mathrm{A}^*(R) = \mathrm{A}_1^*(R)$; for all $R > \hat{R}$ where $\hat{R} > \beta^{-1}$, $\mathrm{A}^*(R) = \mathrm{A}_2^*(R)$; and if $R = \hat{R}$, $\mathrm{A}^*(R) = \mathrm{A}_1^*(R) \cup \mathrm{A}_2^*(R)$. Moreover, $a_2^*(\hat{R}) > p^* > a_1^*(\hat{R})$. This establishes that any selection from A^* is strictly increasing in R. Finally, as $R \to +\infty$, $a_2^* \to +\infty$ and hence $\mathrm{A}^*(R)$ grows unbounded as $R \to +\infty$. ■

Proof of Proposition 2. Existence and uniqueness of the equilibrium rate. The left side of (12), which represents the asset supply valued at the price $q+d$, is strictly decreasing in R. It approaches $+\infty$ as R tends to 1 and it approaches Ad as R tends to $+\infty$. From Proposition 1, any selection from the asset demand correspondence, $\mathrm{A}^d(R)$, is strictly increasing for all R. By (9), $a_1^*(1) \in \mathrm{A}^d(1)$ is finite, and by (10) $a_2^*(+\infty) = +\infty$. Thus, the asset supply is larger than the asset demand as $R \to 1^+$ and the opposite is true when R is sufficiently large. Since $\mathrm{A}^d(R)$ is upper hemicontinuous and convex-valued, any discontinuity in individual asset demand generated by indifference is filled by the aggregate demand correspondence. Hence, a solution to (12) exists and belongs to the interval $(1, +\infty)$. The equilibrium rate is unique because $\mathrm{A}^d(R)$ is ordered monotonically—every selection from $\mathrm{A}^d(R)$ is strictly increasing in R—while the asset-supply schedule is single-valued and strictly decreasing. The argument is illustrated in the left panel of Figure 4. As A increases, the asset supply on the left side of (12) increases. Thus, R increases weakly. Graphically, $A^s(R)$ shifts upward, as shown with the dashed curves in Figure 4. We now turn to different regimes for the equilibrium of the asset market.

Case #1. Define

$$A_0 \equiv \frac{(1-\beta)\,\tilde{a}}{d}.$$

where $\tilde{a} > 0$ is the solution to $\Delta^{b\prime}(\tilde{a}) = 0$. From Proposition 1, $\mathrm{A}^d(\beta^{-1}) = \mathrm{A}_1^*(\beta^{-1}) = \{\tilde{a}\}$. Hence, when $A = A_0$, the asset market clears at $R = \beta^{-1}$. For all $R < \beta^{-1}$, $\max \mathrm{A}^d(R) = \max \mathrm{A}_1^*(R) < \tilde{a}$, where we have used that $\mathrm{A}_1^*(R)$ is a strictly increasing correspondence. Thus, for all $A < A_0$, $R^* < \beta^{-1}$. Using that the asset demand is given by (39) at equality, R^* obeys (13). Given that $a_1^* < \tilde{a}$ for all $a_1^* \in \mathrm{A}_1^*(R)$ and all $R < \beta^{-1}$, $\Delta^{b\prime}(a_1^*) > 0$. Moreover,

$a_1^* < p^*$ implying that buyers' liquidity constraints in the DM are binding.

Case #2. Define

$$A_1 \equiv \frac{\left(\hat{R}-1\right)\min \mathrm{A}_1^*(\hat{R})}{d\hat{R}}.$$

By definition of $\hat{R}$,

$$\min \mathrm{A}_1^*(\hat{R}) \in \mathrm{A}^d(\hat{R}),$$

where we used that $\mathrm{A}_1^*(\hat{R})$ is compact-valued by the Theorem of the Maximum. From Proposition 1, $A^d(\hat{R}) = \text{conv}\left[\mathrm{A}_1^*(\hat{R}) \cup \mathrm{A}_2^*(\hat{R})\right]$. It follows that when $A = A_1$, $R = \hat{R}$ solves (12). Hence, as A increases from A_0 to A_1, R increases from β^{-1} to $\hat{R}$. By the same logic as in Case #1, R^*, solves (39) at equality, which corresponds to (13). Given that $a_1^* \in (\tilde{a}, p^*)$ for all $a_1^* \in \mathrm{A}_1^*(R)$ and for all $R \in (\beta^{-1}, \hat{R})$, $\Delta^{b\prime}(a_1^*) < 0$ and buyers' liquidity constraints in the DM are binding.

Case #3. Define

$$A_2 \equiv \frac{\left(\hat{R}-1\right)a_2^*(\hat{R})}{d\hat{R}}.$$

Since $\hat{R} > \beta^{-1}$ and asset demand is strictly increasing below $\hat{R}$, $\min \mathrm{A}_1^*(\hat{R}) > \tilde{a}$, implying $A_1 > A_0$. Moreover, from Proposition 1, $a_2^*(\hat{R}) > p^* > \min \mathrm{A}_1^*(\hat{R})$, so $A_2 > A_1$. Hence, $A_0 < A_1 < A_2$. By definition of $\hat{R}$, $\mathrm{A}^d(\hat{R}) = \left[\min \mathrm{A}_1^*(\hat{R}), a_2^*\right]$ where a_2^* is the single element in $\mathrm{A}_2^*(\hat{R})$. Thus, when $A = A_2$, the solution to (12) is $R = \hat{R}$. It follows that for all $A \in [A_1, A_2]$, $R = \hat{R}$. Generically, $\mathrm{A}_1^*(\hat{R}) = \{a_1^*\}$. Moreover, from Proposition 1, $\mathrm{A}_2^*(\hat{R}) = \{a_2^*\}$. When $A \in (A_1, A_2)$, in order to achieve market clearing, some buyers must be holding a_1^* while other buyers hold a_2^*. Let ω be the share of buyers holding $a_1^*(\hat{R})$. From (12), it satisfies:

$$\omega a_1^* + (1-\omega)a_2^* = A\left(\frac{d\hat{R}}{\hat{R}-1}\right).$$

Solving for ω we obtain:

$$\omega = \frac{a_2^* - A\left(\frac{d\hat{R}}{\hat{R}-1}\right)}{a_2^* - a_1^*}.$$

Liquidity constraints in the DM only bind for the ω buyers holding a_1^*.

Case #4. For all $A > A_2$, $R > \hat{R}$ and $\mathrm{A}^d(R) = \{a_2^*(R)\}$ where $a_2^*(R)$ is given by (42). By market clearing, $a_2^*(R) = A\left(\frac{dR}{R-1}\right)$, which gives (14). Since $a_2^*(R) > p^*$, liquidity constraints are slack. ■

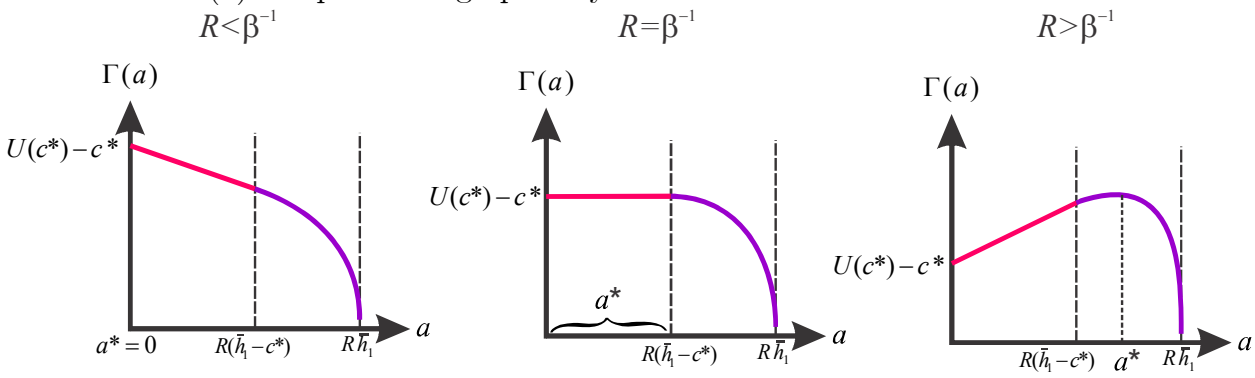
Proof of Lemma 1. When $\alpha = 0$ the optimal choice of asset holdings reduces to

$$\mathrm{A}^d(R) = \arg \max_{a \in \left[0, R\bar{h}_1\right]} \Gamma(a),$$

where

$$\Gamma(a) \equiv \max_{c \geq 0} \left\{ U(c) - c - \left(\frac{1 - \beta R}{R} \right) a \right\} \quad \text{s.t.} \quad c \leq \bar{h}_1 - \frac{a}{R}, \quad \forall a \in \left[0, R\bar{h}_1 \right]. \tag{45}$$

The function $\Gamma(a)$ is represented graphically below.



Function $\Gamma(a)$ for different values of R

The function $\Gamma(a)$ is continuous, differentiable, linear for all $a \leq R(\bar{h}_1 - c^*)$, and strictly concave otherwise. If $\beta R < 1$, then $\Gamma(a)$ is decreasing everywhere and hence $\text{A}^d(R) = \{0\}$. If $\beta R = 1$, $\Gamma(a) = U(c^*) - c^*$ for all $a \leq R(\bar{h}_1 - c^*)$, where $\bar{h}_1 - c^* > 0$ by assumption, and it is strictly decreasing otherwise. In that case, $\text{A}^d(R) = [0, R(\bar{h}_1 - c^*)]$. Finally, if $\beta R > 1$, $\Gamma'(a) = (\beta R - 1)/R > 0$ for all $a \leq R(\bar{h}_1 - c^*)$. Thus, $\text{A}^d(R)$ is given by the unique $a < R\bar{h}_1$ solving the first-order condition

$$U'\left(\bar{h}_1 - \frac{a}{R}\right) = \beta R. \tag{46}$$

The solution is increasing in R. ■

Proof of Proposition 3. From (16), the asset demand correspondence is equal to $\{0\}$ when $R < \beta^{-1}$; to $[0, R(\bar{h}_1 - c^*)]$ when $R = \beta^{-1}$; and to $\left\{R\left[\bar{h}_1 - U'^{-1}(\beta R)\right]\right\}$ when $R > \beta^{-1}$. Hence, any selection from $A^d(R)$ is nondecreasing in R, and strictly increasing when $R > \beta^{-1}$. Moreover, as $R \to +\infty$, $\text{A}^d(R) \to +\infty$. Thus, as illustrated in the left panel of Figure 5, there exists an R solution to the market-clearing condition, (12), i.e.,

$$\frac{AdR}{R-1} \in \text{A}^d(R),$$

and it is unique.

We distinguish two regimes, $R = \beta^{-1}$ and $R > \beta^{-1}$. The first regime, $R = \beta^{-1}$, requires

$$\frac{dA}{1-\beta} \in \text{A}^d(\beta^{-1}) = \left[0, \frac{\bar{h}_1 - c^*}{\beta}\right].$$

This condition is satisfied if and only if $dA/(1-\beta) \leq (\bar{h}_1 - c^*)/\beta$, i.e., (17) holds. From (16) and (12), if $R > \beta^{-1}$, then

$$R\left[\bar{h}_1 - U'^{-1}(\beta R)\right] = A\left(\frac{dR}{R-1}\right).$$

Simplify and rearrange this expression to obtain (18).

We now establish that the threshold for A, $\breve{A}_1$, that separates the two regimes is between A_0 and A_2. From above, this threshold solves $d\breve{A}_1/(1-\beta) = (\bar{h}_1 - c^*)/\beta$, i.e.,

$$\breve{A}_1 = \frac{(1-\beta)(\bar{h}_1 - c^*)}{d\beta}.$$

From the proof of Proposition 2,

$$\begin{aligned} A_0 &= \frac{(1-\beta)\,\tilde{a}}{d}. \\ A_2 &= \frac{\left(\hat{R}-1\right)a_2^*(\hat{R})}{d\hat{R}} \end{aligned}$$

Using that, by Assumption 2, $\tilde{a} < p^* \leq \bar{h}_1 - c^*$, it follows immediately that $A_0 < \breve{A}_1$. Using that $\hat{R} - 1 > \rho = (1-\beta)/\beta$ and $a_2^*(\hat{R})/\hat{R} > \bar{h}_1 - c^*$, it follows that $A_2 > \breve{A}_1$. ■

Proof of Proposition 4. From (20), the value of money in a symmetric equilibrium is a q such that $Aq(1+i_m) \in \text{A}^*(1+i_m)$. From the properties of $\text{A}^*(R)$ described in Proposition 1, $\text{A}^*(R)$ is nonempty, upper hemicontinuous, and any selection from $\text{A}^*(R)$ is increasing in R. Moreover, $\text{A}^*(1+i_m) \subseteq (0, (1+i_m)\bar{h}_1]$. Hence, q exists, is positive and finite, and $Aq(1+i_m)$ is increasing with i_m.

We now turn to the role of taxes in individual money demand. Since $i_m < \bar{\iota}$, $\tau < \bar{h}_2$, so an old agent can pay the tax for any wealth $a \geq 0$ upon entering the CM. Hence, her continuation value is $W_o^\tau(a) = a - \tau$ for every $a \geq 0$. Because the tax enters additively and is independent of the agent's money holdings, it does not affect DM bargaining surpluses. Future labor remains unpledgeable, so the liquidity constraint $p \leq a$ and the terms of trade are unchanged. The tax also does not affect the individual money-demand problem. The demand correspondence is therefore the same as $\text{A}^*(R)$ characterized in Proposition 1.

Finally, the threshold, $\hat{R}$, is defined in Proposition 1. From Proposition 1, for all $i_m \in [0, \hat{R}-1)$, real balances are given by $a_1^* = (1+i_m)qA$, which is a solution to (39) and corresponds to (21). From Proposition 1 and Lemma 4, for all $i_m \in (\hat{R}-1, \bar{\iota})$, real balances are equal to $a_2^* = R\left(\bar{h}_1 - c\right) > p^*$ where from (10), $c < c^*$ is the solution to

$$U'(c) = \beta R = \frac{1+i_m}{1+\rho}.$$

Subtract one from both sides to obtain (22). When $i_m = \hat{R} - 1$, there are two symmetric equilibria, $a_1^* \in \mathrm{A}_1^*(\hat{R})$ and $a_2^* \in \mathrm{A}_2^*(\hat{R})$. Since we assume $\Delta^b(a)$ is strictly concave on $(0, p^*)$, $a_1^* < p^*$ is the unique solution to (21) and $a_2^* = (1 + i_m)(\bar{h}_1 - c) > p^*$, where $c < c^*$ solves (22). ■

Proof of Proposition 5. Part 1: Optimal interest on currency. Since $\bar{\iota} > \hat{\iota}$, all policies in $[0, \hat{\iota}]$ are feasible; on the higher-interest branch we consider only $i_m \in (\hat{\iota}, \bar{\iota})$. From Proposition 4, for all $i_m \in [0, \hat{R} - 1)$, real balances solve

$$\alpha\sigma\Delta^{b\prime}(a^*) = \frac{\rho - i_m}{1 + i_m}.$$

As i_m increases, a^* increases, and $y = w^{-1}(a^*)$ increases. Since $y < y^*$ and $c = c^*$, social welfare increases. For all $i_m \in (\hat{R} - 1, \bar{\iota})$, real balances are equal to $a^* = R(\bar{h}_1 - c) > p^*$ where $c < c^*$ is the solution to

$$U'(c) = (1 + i_m)\beta.$$

As i_m increases, c falls further below c^* while $y = y^*$. Since welfare is increasing in i_m for $i_m < \hat{\iota}$ and decreasing for $i_m > \hat{\iota}$, the optimal interest rate is $i_m = \hat{\iota}$. At $i_m = \hat{\iota} = \hat{R} - 1$, $\mathrm{A}^*(1 + i_m) = \{a_1^*, a_2^*\}$. Therefore, the maximum welfare attainable corresponds to either $a^* = a_1^*$ or a_2^*. In any equilibrium when $i_m = \hat{\iota}$, either agents consume $y < y^*$ and $c = c^*$ or they consume $y = y^*$ and $c < c^*$. Thus, the optimal policy fails to implement the first best.

Part 2: Optimality of the Friedman rule at the limit. Consider a sequence of economies in which $\bar{h}_1 \to +\infty$ and $\bar{h}_2$ increases as needed so that Assumption 3 continues to hold. Recall that the threshold, $\hat{R}$, solves:

$$\begin{aligned} & U(c^*) - c^* + \beta \max_{a \in [0, p^*]} \left\{ - \left(\frac{1 - \beta\hat{R}}{\beta\hat{R}} \right) a + \alpha\sigma\Delta^b(a) \right\} \\ = & \max_{c \leq \bar{h}_1 - p^*/\hat{R}} \left\{ U(c) - \beta\hat{R}c \right\} + \left(\beta\hat{R} - 1 \right) \bar{h}_1 + \beta\alpha\sigma\Delta^b(p^*). \end{aligned}$$

Using that $\hat{R} \geq \beta^{-1}$ and $\hat{R}$ is decreasing in $\bar{h}_1$, $\lim_{\bar{h}_1 \to +\infty} \hat{R}$ exists. Suppose $\lim_{\bar{h}_1 \to +\infty} \hat{R} > \beta^{-1}$. Then the right side tends to $+\infty$ as $\bar{h}_1 \to +\infty$ whereas the left side is bounded. A contradiction. Hence, for the equality to hold, it must be that $\hat{R} \to \beta^{-1} = 1 + \rho$. On the right side, consumption approaches c^* and $a = \hat{R}(\bar{h}_1 - c) > p^*$ so that $y = y^*$. Thus, as $\bar{h}_1 \to +\infty$, $\hat{\iota} \searrow \rho$ and $(y, c) \longrightarrow (y^*, c^*)$ along the high-balance equilibrium, $a^* = a_2^*$. Hence, the optimal policy implements an allocation approaching the first best. ■

Proof of Lemma 2. The value of an agent in the DM can be reexpressed as follows:

$$V(\mathbf{a}) = \sum_{j \in \mathbb{J}} \alpha_j \sigma \Delta^b(a^j) + \alpha\sigma\mathbb{E}(\Delta^s) + \mathbf{a}.\mathbf{1} + W_o(0), \tag{47}$$

where $\mathbf{a} \equiv (a^j)$ is the portfolio (vector) of assets and $\mathbf{1} \equiv (1, ..., 1)^\top$. The first term on the right side corresponds to the buyer's expected DM surplus. A type-j meeting occurs with probability α_j, in which case only type-j assets are accepted and enter the surplus function. The second term is the agent's expected DM surplus as a seller. The last two terms correspond to the continuation value, $W_o(\mathbf{a})$, which is linear with a positive intercept due to the assumption that old agents receive the newly created assets.

The value function of a young agent is given by

$$W_y = \max_{c,h,\mathbf{a}} \{U(c) - h + \beta V(\mathbf{a})\} \quad \text{s.t.} \quad c + \mathbf{a}.\mathbf{q} = h \text{ and } h \leq \bar{h}_1,$$

where $\mathbf{q} \equiv (q^j)^\top$. The agent chooses her portfolio, $\mathbf{a}$, her labor supply, h, and her consumption, c, to maximize her CM utility subject to a budget constraint. Replace $V(\mathbf{a})$ with its expression given by (47) to obtain

$$W_y = \max_{\mathbf{a} \in \mathbb{R}_+^{|\mathbb{J}|}} \left\{\Gamma(\mathbf{a}) : \mathbf{a}.\mathbf{q} \leq \bar{h}_1\right\} + \beta \left[\alpha\sigma\mathbb{E}\left(\Delta^s\right) + W_o(0)\right], \tag{48}$$

where Γ is defined in (25). ■

Proof of Proposition 6. Fix a finite vector of positive asset supplies $\mathbf{A}$. For every strictly positive price vector $\mathbf{q}$, the budget set in (24) is compact and $\Gamma(\mathbf{a})$ is continuous. Hence, $\mathbf{A}^*(\mathbf{q})$ is nonempty and compact valued and, by Berge's maximum theorem, upper hemicontinuous. It follows that $\text{A}^d(\mathbf{q}) = \text{conv}[\mathbf{A}^*(\mathbf{q})]$ is nonempty, compact, convex valued, and upper hemicontinuous. Prices can be restricted to a compact rectangle. An equilibrium cannot have $q^j > \bar{h}_1/A^j$ because every feasible portfolio satisfies $a^j \leq \bar{h}_1/q^j < A^j$, so market clearing for asset j would be impossible. A zero price cannot be an equilibrium either because every asset delivers a strictly positive payoff in the next CM and would therefore be in excess demand. Hence, all equilibrium prices belong to $\prod_{j\in\mathbb{J}}[\varepsilon, \bar{h}_1/A^j]$ for some sufficiently small $\varepsilon > 0$. Given these properties, existence follows from the standard general-equilibrium argument based on Kakutani's fixed-point theorem; see, e.g., Mas-Colell, Whinston, and Green (1995, Chapter 17). ■

Lemma 5 *For given prices, $\mathbf{q} \in \mathbb{R}_{++}^{|\mathbb{J}|}$, the optimal asset holdings solve*

$$-q^j U'(c) + \beta \left[1 + \alpha_j \sigma \Delta^{b\prime}(a^j)\right] \leq 0, \tag{49}$$

with equality if $a^j > 0$, for all $j \in \mathbb{J}$, and where $c = \min\{c^*, \bar{h}_1 - \mathbf{a}.\mathbf{q}\}$.

Proof of Lemma 5. The function Γ is differentiable everywhere except at $\mathbf{a}$'s such that $\alpha_j > 0$ and $a^j = p^*$. (If $\alpha_j = 0$, the kink at p^* is irrelevant because the term involving $\Delta^b(a^j)$ drops out of Γ). We now establish that these points are never a solution to $\max_{\mathbf{a}} \Gamma(\mathbf{a})$. The argument goes as follows. Suppose $\alpha_j > 0$ and the optimal solution is such that $a^j = p^*$. Since $U'(0) = +\infty$, any optimum has $c > 0$ so that the small variations in a^j considered next are feasible. The agent has no incentive to increase a^j by an infinitesimal amount only if the right derivative is non-positive, i.e.,

$$-q^j U'(c) + \beta \leq 0.$$

where we used that $\Delta^{b\prime}(p^{*+}) = 0$. She has no incentive to reduce a^j by an infinitesimal amount only if the left derivative is non-negative,

$$-q^j U'(c) + \beta \left[1 + \alpha_j \sigma \Delta^{b\prime}(p^{*-})\right] \geq 0.$$

Thus, $a^j = p^*$ is optimal only if

$$-q^j U'(c) + \beta \leq 0 \leq -q^j U'(c) + \beta + \beta \alpha_j \sigma \Delta^{b\prime}(p^{*-}),$$

which is inconsistent with $\Delta^{b\prime}(p^{*-}) < 0$. Therefore, we can use the differentiability of $\Gamma(\mathbf{a})$ and, from (25), the first-order necessary conditions are (49). ■

Proof of Proposition 7.

Part 1. We construct a symmetric equilibrium where the constraint, $h \leq \bar{h}_1$, is not binding. Thus, $c = c^*$ and, from market clearing, $a^0 = A^0$ and $a^1 = A^1$. We conjecture and verify later that for such an equilibrium to exist, $A^1 < p^*$. From (49) at equality asset prices solve (28)-(29). We define

$$\underline{q}^1 = \max\{\beta \left[1 + \alpha_1 \sigma \Delta^{b\prime}(p^{*-})\right], 0\}.$$

If $\beta \left[1 + \alpha_1 \sigma \Delta^{b\prime}(p^{*-})\right] > 0$, it corresponds to the price of asset 1 as $A^1 \nearrow p^*$. From (25), the condition for $c = c^*$ is $c^* \leq \bar{h}_1 - \mathbf{a}.\mathbf{q}$ with $a^0 = A^0$, $a^1 = A^1$, and prices given by (28)-(29), which can be rewritten as

$$\bar{h}_1 \geq c^* + \beta A^0 + \beta \left[1 + \alpha_1 \sigma \Delta^{b\prime}(A^1)\right] A^1. \tag{50}$$

Under (27) this condition holds. Under the assumption that $\Delta^b(a)$ is strictly concave on $(0, p^*)$, $a^1 = A^1$ is the unique solution to $\max_{\mathbf{a}} \Gamma(\mathbf{a})$ when a^1 is restricted to $[0, p^*]$. We now rule out a solution to $\max_{\mathbf{a}} \Gamma(\mathbf{a})$ such that $a^1 \geq p^*$. From (25), such an alternative solution is given by

$$\breve{a}^1 = \arg \max_{a^1 \geq p^*, c \geq 0} \left\{U(c) - c - a^1(q^1 - \beta) : c + a^1 q^1 \leq \bar{h}_1\right\}, \tag{51}$$

where we used that since $q^0 = \beta$, $a^0 = 0$ with no loss in generality. If $q^1 \geq \beta$, we can select $\breve{a}^1 = p^*$ as an optimum, which is not a profitable deviation relative to the optimal a^1 (since a^1 is allowed to be equal to p^*). Thus, in the following, we focus on the case $q^1 \in \left(\underline{q}^1, \beta\right)$. At the optimum of (51), the constraint $c + a^1 q^1 \leq \bar{h}_1$ binds and $c = \breve{c} = \bar{h}_1 - q^1 \breve{a}^1$. The portfolio with $a^1 \leq p^*$ weakly dominates the portfolio with $\breve{a}^1 \geq p^*$ if $\Psi(q^1) \geq 0$ where

$$\begin{aligned} \Psi(q^1) \equiv & \max_{a^1 \leq p^*, c + a^1 q^1 \leq \bar{h}_1} \left\{ U(c) - c - a^1 (q^1 - \beta) + \beta \alpha_1 \sigma \Delta^b(a^1) \right\} \\ - & \max_{p^* \leq \breve{a}^1 \leq \bar{h}_1 / q^1} \left\{ U(\bar{h}_1 - q^1 \breve{a}^1) - \bar{h}_1 + \breve{a}^1 \beta + \beta \alpha_1 \sigma \Delta^b(p^*) \right\}. \end{aligned} \tag{52}$$

By construction, the optimal $a^1 \leq p^*$ is such that $c + a^1 q^1 \leq \bar{h}_1$ is slack and $c = c^*$. The function $\Psi(q^1)$ is continuous in q^1, strictly increasing for all $q^1 < \beta$ with

$$\Psi'(q^1) = -a^1 + U'(\breve{c}) \breve{a}^1 > 0.$$

If $\underline{q}^1 > 0$, as q^1 approaches $\underline{q}^1$, a^1 approaches p^* and $\Psi(q^1) < 0$. If $\underline{q}^1 = 0$, then $\breve{a}^1 = +\infty$ and as q^1 approaches $\underline{q}^1$, $\Psi(q^1)$ tends to $-\infty$. Moreover, $\Psi(\beta) = \beta \alpha_1 \sigma \left[\Delta^b(\tilde{a}) - \Delta^b(p^*) \right] > 0$. Thus, there is a unique solution $\hat{q}^1 \in (\underline{q}^1, \beta)$ to $\Psi(\hat{q}^1) = 0$. We denote $\underline{A}^1 \in (\tilde{a}, p^*)$ the value of A^1 such that $q^1 = \hat{q}^1$, i.e.

$$\hat{q}^1 = \beta \left\{ 1 + \alpha_1 \sigma \Delta^{b\prime}(\underline{A}^1) \right\}.$$

For all $A^1 \leq \underline{A}^1$, $q^1 \geq \hat{q}^1$, and $\Psi(q^1) \geq 0$.

Part 2. Suppose that $A^1 > \underline{A}^1$. We construct asymmetric equilibria where $q^1 = \hat{q}^1$. Suppose a fraction $\omega > 0$ of agents hold the two assets and their labor supply constraint is not binding so that $c = c^*$. From the first-order conditions (49) at equality, $q^0 = \beta$ and $a^1 = \underline{A}^1$. The remaining $1 - \omega$ agents only hold the liquid asset and their choice is given by $\breve{a}^1$ solution to (51). The market clearing condition for the liquid asset is:

$$\omega \underline{A}^1 + (1 - \omega) \breve{a}^1 = A^1. \tag{53}$$

Thus, (53) determines ω, which can be solved in closed form to obtain (32). Since $\underline{A}^1 < \breve{a}^1$, the condition $A^1 > \underline{A}^1$ guarantees that $\omega < 1$. The labor supply constraint of the first type of agents does not bind if:

$$\bar{h}_1 \geq c^* + q^0 \frac{A^0}{\omega} + q^1 a^1 = c^* + \beta \frac{A^0}{\omega} + \hat{q}^1 \underline{A}^1. \tag{54}$$

After substituting ω given by (53), i.e., $\omega = \left(\breve{a}^1 - A^1\right) / \left(\breve{a}^1 - \underline{A}^1\right)$, into (54) at equality we obtain:

$$\bar{h}_1 = c^* + \frac{\beta A^0 \left(\breve{a}^1 - \underline{A}^1\right)}{\breve{a}^1 - A^1} + \hat{q}^1 \underline{A}^1.$$

The solution for A^1, denoted $\bar{A}^1$, is an upper bound above which (54) binds. It solves:

$$\bar{A}^1 = \breve{a}^1 - \frac{\beta A^0 \left(\breve{a}^1 - \underline{A}^1\right)}{\bar{h}_1 - c^* - \hat{q}^1 \underline{A}^1}. \tag{55}$$

It is the largest asset supply that can be absorbed while keeping the labor supply constraint of the first type of agents slack. Under condition (27), $\bar{h}_1 > c^* + \beta A^0 + \hat{q}^1 \underline{A}^1$. Thus, $\bar{A}^1 > \underline{A}^1$. Moreover, $\bar{A}^1 < \breve{a}^1$ so that $\omega > 0$ at $A^1 = \bar{A}^1$.

Part 3. Suppose $A^1 > \bar{A}^1$. Existence follows from Proposition 6. We now characterize any equilibrium. First, we establish that $q^0 < \beta$. The proof is by contradiction. Suppose $q^0 = \beta$. Thus, the labor supply constraint of agents holding the illiquid asset is slack. If $q^1 > \hat{q}^1$, then the portfolio $(0, \breve{a}^1)$ is strictly dominated by the portfolio involving both assets. Hence, any equilibrium must be of the form described in part 1. From part 1, $q^1 > \hat{q}^1$ requires $A^1 < \underline{A}^1$, which contradicts our assumption $A^1 > \bar{A}^1 > \underline{A}^1$. If $q^1 = \hat{q}^1$, and given that $A^1 > \bar{A}^1$, then the labor supply constraint, (54), cannot hold. Thus, it must be that $q^1 < \hat{q}^1$. We have shown in part 1 that if $q^1 < \hat{q}^1$ all agents want to hold $\breve{a}^1$ units of the liquid asset and no units of the illiquid asset. Hence, markets do not clear. This contradiction establishes $q^0 < \beta$.

Second, we show that $q^1 < q^0$. Suppose, by contradiction, that $q^1 \geq q^0$. Since aggregate liquid-asset supply exceeds $\tilde{a}$, some agents must hold $a^1 > \tilde{a}$ units of the liquid asset. But since $q^1 \geq q^0$, such agents could reduce a^1 toward $\tilde{a}$ and replace the portfolio value with weakly cheaper illiquid assets, thereby increasing their DM surplus while leaving wealth unchanged, a contradiction. Formally, the surplus from the deviation that consists in reducing a^1 to $\tilde{a}$ and raising a^0 to $a^0 + (a^1 - \tilde{a})q^1/q^0$ so that c is constant is equal to

$$\begin{aligned}
&U(c) - c - \tilde{a}(q^1 - \beta) - \left[a^0 + (a^1 - \tilde{a})\frac{q^1}{q^0}\right](q^0 - \beta) + \beta\alpha_1\sigma\Delta^b(\tilde{a}) \\
&= U(c) - c - \left(a^0 q^0 + a^1 q^1\right) + \beta\tilde{a} + \beta\left[a^0 + (a^1 - \tilde{a})\frac{q^1}{q^0}\right] + \beta\alpha_1\sigma\Delta^b(\tilde{a}) \\
&> U(c) - c - \left(a^0 q^0 + a^1 q^1\right) + \beta\left(a^0 + a^1\right) + \beta\alpha_1\sigma\Delta^b(a^1).
\end{aligned}$$

Thus, the deviation is profitable.

Finally, we establish that the liquidity constraint, $p \leq a^1$, binds for some agents. The proof is by contradiction. Suppose that no agent is liquidity constrained, i.e., all agents hold $a^1 \geq p^*$ and some agents hold the illiquid asset since $A^0 > 0$. Optimality then implies that the two assets have the same price, $q^1 = q^0$. This contradicts the result above that $q^1 < q^0$.

■

Proof of Proposition 8. Part 1: Conventional yield curve. We conjecture that the equilibrium prices are given by (33) and $q^0 = \beta$, and we verify that under the conditions

in the proposition, these prices are indeed part of an equilibrium. From (34) and (33), $q^2 > q^1 > \beta$. From the first-order necessary conditions, (49), if $q^j > \beta$, any optimal solution must be such that $a^j \in [0, \tilde{a}]$. Indeed, since $c \leq c^*$, $U'(c) \geq U'(c^*) = 1$ and $-q^j U'(c) + \beta < 0$, which implies $\Delta^{b\prime}(a^j) > 0$ and hence $a^j < \tilde{a}$. By the strict concavity of Δ^b over that interval, a^j is uniquely determined by (49) and from (33) $a^j = A^j$ is the solution. Thus, q^j is a market-clearing price. From (49) at equality with $\alpha_0 = 0$ and $c = c^*$, $q^0 = \beta$. Thus, $a^0 = A^0$ is an optimal choice and $q^0 = \beta$ is a market-clearing price. Under the condition $\bar{h}_1 \geq c^* + \mathbf{A}.\mathbf{q}$, c^* is feasible. Thus, we have checked that asset prices given by (33) and $c = c^*$ are part of an equilibrium.

Part 2: Partial inversion. We show that, given the conditions on asset supplies, the prices given by (33) are part of a symmetric equilibrium. The assumption $A^j < \tilde{a}$ implies $q^j > \beta$ and the assumption $A^{j'} \in (\tilde{a}, \breve{a})$ implies $q^{j'} < \beta$ provided $\breve{a} < p^*$. Thus, prices are ranked as stated in the proposition, $q^j > q^0 = \beta > q^{j'}$. We now check that $a^j = A^j$ are optimal choices for all $j \in \{0, 1, 2\}$ so that markets clear. First, $a^j = A^j$ satisfies the necessary conditions for optimality, (49), when $c = c^*$. For sufficiency, we need to check there are no profitable deviations. Denote $\hat{\mathbf{a}}$ an alternative portfolio that also satisfies (49). The alternative portfolio must feature either $\hat{a}^1 > p^*$ or $\hat{a}^2 > p^*$ since, by the strict concavity of $\Delta^b(a)$, there is a unique solution (a^1, a^2) to (49) on $[0, p^*]^2$ and it corresponds to (A^1, A^2). Given that $q^j > \beta$, (49) implies $\Delta^{b\prime}(\hat{a}^j) > 0$, i.e., $\hat{a}^j < \tilde{a}$. It follows that if there is a profitable deviation, it must feature $\hat{a}^{j'} > p^*$. Consider such a deviation, in which case $\Delta^{b\prime}(\hat{a}^{j'}) = 0$. From (49) at equality and $\beta > q^{j'}$, it follows that $\hat{c} < c^*$. Thus, from (49) when $j = 0$, $\hat{a}^0 = 0$, and $(\hat{a}^j, \hat{a}^{j'})$ solve

$$q^j = q^{j'}\left[1 + \alpha_j \sigma \Delta^{b\prime}(\hat{a}^j)\right] \tag{56}$$

$$U'(\hat{c}) = \frac{\beta}{q^{j'}}, \tag{57}$$

where $\hat{c} = \bar{h}_1 - \hat{a}^j q^j - \hat{a}^{j'} q^{j'}$. This deviation is not profitable if $\Gamma(\mathbf{a}) \geq \Gamma(\hat{\mathbf{a}})$, i.e.,

$$\begin{aligned} & U(c^*) - c^* - \mathbf{A}.(\mathbf{q} - \boldsymbol{\beta}) + \beta \sum_{j \in \{1,2\}} \alpha_j \sigma \Delta^b(A^j) \\ \geq\ & U(\hat{c}) - \hat{c} - \hat{\mathbf{a}}.(\mathbf{q} - \boldsymbol{\beta}) + \beta \alpha_j \sigma \Delta^b(\hat{a}^j) + \beta \alpha_{j'} \sigma \Delta^b(p^*). \end{aligned} \tag{58}$$

The left side is the utility from the candidate portfolio while the right side is the utility from the alternative portfolio given by (56)-(57). Suppose $A^{j'} = \tilde{a}$. Then, from (33) $q^{j'} = \beta$ and $\Delta^b(A^{j'}) > \Delta^b(p^*)$. From (56)-(57) $\hat{a}^j = A^j$, $\hat{c} = c^*$, and $\mathbf{a}.(\mathbf{q} - \boldsymbol{\beta}) = \hat{\mathbf{a}}.(\mathbf{q} - \boldsymbol{\beta})$. Therefore, the condition (58) holds as a strict inequality. By continuity, there is a non-empty interval $(\tilde{a}, \breve{a})$ such that for all $A^{j'} \in (\tilde{a}, \breve{a})$ the inequality holds. Note that for this argument we do allow $\breve{a}$ to be a function of A^j. Finally, to guarantee that $c = c^*$, $\bar{h}_1 \geq c^* + \mathbf{A}.\mathbf{q}$.

Part 3: Full inversion. We conjecture that the equilibrium prices are given by (33) and $q^0 = \beta$, and verify that these prices are part of a symmetric equilibrium when A^1 and A^2 are sufficiently close to $\tilde{a}$. Under (35), (33) implies

$$\beta > q^1 > q^2.$$

Moreover, $a^j = A^j$, $j \in \{1, 2\}$, together with $c = c^*$, satisfies the first-order necessary conditions (49).

We now establish sufficiency. By strict concavity of $\Delta^b(a)$ on $[0, p^*]$, the first-order conditions have a unique solution on $[0, p^*]^2$, namely (A^1, A^2). Hence, as in Part 2, any alternative candidate optimum must have $\hat{a}^1 > p^*$ or $\hat{a}^2 > p^*$.

Consider first the limiting supplies $A^1 = A^2 = \tilde{a}$. From (33),

$$q^1 = q^2 = q^0 = \beta.$$

At these prices, the objective in (25) can be written as

$$\Gamma(\mathbf{a}) = U(c) - c + \beta \sum_{j \in \{1,2\}} \alpha_j \sigma \Delta^b(a^j),$$

where $c \leq c^*$. Since c^* maximizes $U(c) - c$ and $\tilde{a}$ strictly maximizes $\Delta^b(a)$, the portfolio with $a^1 = a^2 = \tilde{a}$ and $c = c^*$ is optimal. Moreover, any portfolio with $\hat{a}^j > p^*$ for some $j \in \{1, 2\}$ is strictly dominated, because

$$\Delta^b(\hat{a}^j) = \Delta^b(p^*) < \Delta^b(\tilde{a}).$$

Thus, at $A^1 = A^2 = \tilde{a}$, all candidate deviations outside $[0, p^*]^2$ yield strictly lower utility than the candidate equilibrium portfolio.

The objective Γ, the prices in (33), and the candidate portfolio vary continuously with (A^1, A^2). Moreover, for prices in a neighborhood of β, the budget constraint implies that the set of feasible portfolios is bounded. Therefore, the strict inequality established above is preserved for all (A^1, A^2) in a sufficiently small neighborhood of $(\tilde{a}, \tilde{a})$. Hence, there exists $\breve{a}(\bar{h}_1) > \tilde{a}$ such that, for all

$$(A^1, A^2) \in (\tilde{a}, \breve{a})^2,$$

no deviation with $\hat{a}^1 > p^*$ or $\hat{a}^2 > p^*$ is profitable. Together with the uniqueness of the solution to the first-order conditions on $[0, p^*]^2$, this establishes that $a^j = A^j$ is globally optimal. Finally, for $\bar{h}_1$ sufficiently large, $c = c^*$ is feasible. Thus, whenever (35) holds, the prices in (33) are equilibrium prices and satisfy

$$q^0 = \beta > q^1 > q^2.$$

■

SUPPLEMENTARY ONLINE APPENDICES

The supplementary online appendices provide additional results, foundations, and extensions of the analysis in the paper. They are organized as follows.

B1. Concavity of $\Delta^b(a)$. We provide a sufficient condition for the buyer's trade surplus to be strictly concave under CRRA preferences, thereby providing a simple sufficient condition for Assumption 1a.

B2. Foundations for the Nash solution. We provide strategic foundations for the generalized Nash solution using an infinite-horizon alternating-offer bargaining game and characterize how the buyer's endogenous bargaining share varies with liquidity.

B3. A finite-horizon bargaining game. We consider a simple two-round bargaining game and show that the non-monotonicity of the buyer's trade surplus is not specific to the generalized Nash solution.

B4. Welfare measures and optimal monetary policy. We consider an alternative, Golden Rule welfare criterion and revisit the optimal provision of liquidity under this criterion.

B5. Concealable and segregable asset holdings. We relax the assumption that asset holdings are fully observable and cannot be segregated, and characterize the implications for negative liquidity premia.

B6. Collateralized borrowing with rehypothecation. We introduce collateralized borrowing and lending when collateral can be rehypothecated and show that these arrangements are neutral for equilibrium allocations and asset prices.

B7. Nested acceptability. We consider an alternative multiple-asset environment in which liquid assets compete directly as means of payment and differ in their acceptability across meetings.

B8. Sampling uncertainty in the calibration. We describe the bootstrap procedure used to quantify sampling uncertainty in the calibration, including the construction of the pointwise confidence band for convenience yields and the interval for the zero-crossing threshold.

B1. Concavity of $\Delta^b(a)$

In several parts of the paper, we assume the buyer's trade surplus, $\Delta^b(a)$, is strictly concave for all $a \in (0, p^*)$. In this appendix, we show that this assumption holds for CRRA preferences by providing a simple sufficient condition under which $\Delta^b(a)$ is strictly concave on the constrained region $(0, p^*)$. Also, because strict concavity implies strict unimodality, the result provides an easily verifiable sufficient condition for Assumption 1a.

Recall that under generalized Nash bargaining, the solution to the DM negotiation when the liquidity constraint, $p \leq a$, binds is given by

$$a = w(y;\theta) \equiv \frac{\theta u'(y)y + (1-\theta)u(y)}{\theta u'(y) + (1-\theta)}, \quad \theta \in (0,1). \tag{59}$$

The unconstrained solution is $y = y^*$, where $u'(y^*) = 1$, and $p = p^* \equiv w(y^*;\theta)$. On the constrained region, $a \leq p^*$, $\Delta^b(a) = u\left[y(a)\right] - a$.

Proposition 10 *Suppose $u(y) = y^{1-\gamma}/(1-\gamma)$ with $0 < \gamma < 1$. Then:*

1. $\Delta^{b\prime}(0^+) = +\infty$;
2. $\Delta^b(a)$ *is strictly concave on* $(0, p^*)$ *if*

$$0 < \gamma \leq \frac{1}{2(1-\theta)}. \tag{60}$$

Proof. Part 1: $\Delta^{b\prime}(0^+) = +\infty$. Under the CRRA specification, $u'(y) = y^{-\gamma}$, $u''(y) = -\gamma y^{-\gamma-1} < 0$, and $y^* = 1$. Substituting $u(y) = y^{1-\gamma}/(1-\gamma)$ into (59) gives

$$w(y) = \left(\frac{1-\theta\gamma}{1-\gamma}\right) \frac{y}{\theta + (1-\theta)y^\gamma} \quad \forall y \in [0, y^*].$$

Differentiating,

$$w'(y) = \left(\frac{1-\theta\gamma}{1-\gamma}\right) \frac{\theta + (1-\theta)(1-\gamma)y^\gamma}{\left[\theta + (1-\theta)y^\gamma\right]^2} \quad \forall y \in (0, y^*).$$

Using $0 < \gamma < 1$ and $\theta \in (0,1)$, $w'(y) > 0$ for all $y \in (0, y^*)$. Since $w(0) = 0$ and $w(y^*) = w(1) = p^*$, w maps $(0,1)$ one-to-one onto $(0, p^*)$. Hence, $y(a) = w^{-1}(a)$ is well defined for all $a \in (0, p^*)$. Differentiate the buyer's surplus, $\Delta^b(a) = u\left[w^{-1}(a)\right] - a$, to obtain

$$\Delta^{b\prime}(a) = \frac{u'\left[y(a)\right]}{w'\left[y(a)\right]} - 1.$$

Substituting $u'(y) = y^{-\gamma}$ and the expression for $w'(y)$ gives

$$\Delta^{b\prime}(a) = \left(\frac{1-\gamma}{1-\theta\gamma}\right) \frac{\left[\theta + (1-\theta)y(a)^\gamma\right]^2}{y(a)^\gamma\left[\theta + (1-\theta)(1-\gamma)y(a)^\gamma\right]} - 1. \tag{61}$$

As $a \to 0$, $y(a) \to 0$, hence $y(a)^\gamma \to 0$. It follows that $\lim_{a\to 0^+} \Delta^{b\prime}(a) = +\infty$.

Part 2: Strict concavity of $\Delta^b(a)$. We adopt the change of variable $z \equiv y(a)^\gamma$ and define

$$f(z) \equiv \Delta^{b\prime}\left[a(z)\right] = \left(\frac{1-\gamma}{1-\theta\gamma}\right) \frac{\left[\theta + (1-\theta)z\right]^2}{z[\theta + (1-\theta)(1-\gamma)z]} - 1 \quad \forall z \in (0,1).$$

Since $y(a) \in (0, y^*)$, where $y^* = 1$, $z \in (0,1)$. Differentiating $f(z)$ yields

$$f'(z) = -\frac{\theta(1-\gamma)\left[\theta + (1-\theta)z\right]\left[\theta + (1-\theta)(1-2\gamma)z\right]}{z^2(1-\theta\gamma)\left[\theta + (1-\theta)(1-\gamma)z\right]^2} \quad \forall z \in (0,1).$$

Hence, $f'(z) < 0$ if and only if

$$\theta + (1-\theta)(1-2\gamma)z > 0. \tag{62}$$

Because the expression, $\theta + (1-\theta)(1-2\gamma)z$, is affine in z, its minimum over $[0,1]$ is attained at one of the endpoints. As $z \to 0$, it converges to $\theta > 0$. Hence, a sufficient condition for (62) to hold for all $z \in (0,1)$ is that the expression is nonnegative at $z = 1$, i.e.,

$$\gamma \leq \frac{1}{2(1-\theta)}. \tag{63}$$

By the chain rule,

$$f'(z) = \Delta^{b\prime\prime}\left[a(z)\right] a'(z),$$

where $a(z) \equiv w(z^{1/\gamma})$ and

$$a'(z) = w'(z^{1/\gamma})\frac{1}{\gamma} z^{1/\gamma - 1} > 0.$$

Thus, $\Delta^{b\prime\prime}\left[a(z)\right]$ has the same sign as $f'(z)$, i.e., $\Delta^b(a)$ is strictly concave on $(0, p^*)$ if (63) holds. Hence, condition (63) is a simple sufficient condition for strict concavity of the buyer's trade surplus. ■

In Section 5, we require $\lim_{a\to 0} \Delta^{b\prime}(a)a = 0$. We now establish that this condition holds for CRRA preferences when $\gamma \in (0,1)$. From (61),

$$\Delta^{b\prime}(a)a = \left(\frac{1-\gamma}{1-\theta\gamma}\right) \frac{\left[\theta + (1-\theta)y(a)^\gamma\right]^2 a}{y(a)^\gamma\left[\theta + (1-\theta)(1-\gamma)y(a)^\gamma\right]} - a.$$

Using that $w(y) = a$, we have

$$a = w(y) = \left(\frac{1-\theta\gamma}{1-\gamma}\right) \frac{y}{\theta + (1-\theta)y^\gamma}.$$

Substitute this expression into $\Delta^{b\prime}(a)a$ so that $\lim_{a\to 0} \Delta^{b\prime}(a)a = 0$ if and only if

$$\lim_{y\to 0} \frac{\left[\theta + (1-\theta)y^\gamma\right]^2}{y^\gamma\left[\theta + (1-\theta)(1-\gamma)y^\gamma\right]} \frac{y}{\theta + (1-\theta)y^\gamma} = 0.$$

Simplify this expression to rewrite it as

$$\lim_{y \to 0} \frac{[\theta + (1-\theta)y^{\gamma}]}{[\theta + (1-\theta)(1-\gamma)y^{\gamma}]} y^{1-\gamma} = 0.$$

Hence, the condition $\lim_{a \to 0} \Delta^{b'}(a)a = 0$ holds for all $\gamma \in (0, 1)$.

B2. Foundations for the Nash solution: An infinite-horizon, alternating-offer bargaining game

This appendix provides strategic foundations for the generalized Nash solution used in the text. More importantly, it shows that the buyer's endogenous bargaining share depends on the tightness of the liquidity constraint, thereby explaining the source of the non-monotonicity of the buyer's trade surplus.

We describe succinctly an alternating-offer bargaining game à la Rubinstein (1982) that provides strategic foundations for the solution used in the main text. Because the bargaining takes place within a period, we assume no discounting. Instead, we introduce an exogenous risk of breakdown between each round. Following a round where the seller (respectively, buyer) has rejected an offer, the bargaining continues with probability $e^{-\delta_s \Delta}$ (respectively, $e^{-\delta_b \Delta}$), where $\Delta > 0$ is interpreted as the length of a negotiation round. Later, we will take the limit as Δ goes to 0. Also, even though we allow for an infinite number of rounds, it is well known that the solution to the alternating-offer bargaining game with a finite number of rounds approaches that of the game with infinite horizon as the number of rounds grows unbounded.

We assume the liquidity constraint binds in each round so that effectively the two agents bargain over the output y to be obtained in exchange for the buyer's a units of asset. We focus on stationary strategies whereby each player makes the same offer whenever it is her turn to make an offer and uses the same acceptance rule whenever it is her turn to accept or reject an offer. We denote y^b_Δ the buyer's offer and y^s_Δ the seller's offer. The reservation surplus of the buyer, below which she rejects an offer by the seller, is R^b_Δ. The reservation surplus of the seller is R^s_Δ. Since agents cannot commit, the notion of equilibrium is subgame perfection.

The optimal offer of the buyer makes the seller indifferent between accepting or rejecting,

$$-y^b_\Delta + a = R^s_\Delta = e^{-\delta_s \Delta} \left[-y^s_\Delta + a \right]. \tag{64}$$

On the right side, the reservation surplus of the seller corresponds to her payoff if she has the opportunity to make a counter-offer with probability $e^{-\delta_s \Delta}$. Similarly, the optimal offer of the seller makes the buyer indifferent between accepting and rejecting and is given by

$$u(y^s_\Delta) - a = R^b_\Delta = e^{-\delta_b \Delta} \left[u(y^b_\Delta) - a \right]. \tag{65}$$

From (64), the difference between the buyer's and seller's offers is

$$y^b_\Delta - y^s_\Delta = \left(1 - e^{-\delta_s \Delta} \right) \left[-y^s_\Delta + a \right]. \tag{66}$$

When Δ is small, the buyer raises her demand relative to the seller's offer by a fraction $\delta_s\Delta$ of the seller's surplus. From (65), if we use the approximation $u(y^b_\Delta) \approx u(y^s_\Delta)+(y^b_\Delta - y^s_\Delta)u'(y^s_\Delta)$,

$$y^b_\Delta - y^s_\Delta \approx \frac{\left(1-e^{-\delta_b\Delta}\right)}{u'(y^s_\Delta)}\left[u(y^b_\Delta)-a\right]. \tag{67}$$

The seller lowers her offer by a fraction $\delta_b\Delta/u'(y^s_\Delta)$ of the buyer's surplus. It is as if the effective discount rate of the buyer in the negotiation is scaled down by a factor $u'(y^s_\Delta) > 1$. That factor represents the tightness of the liquidity constraint. The tighter the constraint, the more patient the buyer is in waiting to make a counteroffer.

As $\Delta \to 0$, y^b_Δ and y^s_Δ approach the same y. Rearranging (65) and (64), taking their ratio, and letting Δ go to 0,

$$\lim_{\Delta\to 0}\frac{u(y^b_\Delta)-u(y^s_\Delta)}{y^b_\Delta - y^s_\Delta} = u'(y) = \lim_{\Delta\to 0}\frac{1-e^{-\delta_b\Delta}}{1-e^{-\delta_s\Delta}}\frac{u(y^b_\Delta)-a}{-y^s_\Delta + a} = \frac{\delta_b}{\delta_s}\frac{u(y)-a}{-y+a}.$$

Hence, the ratio of the buyer's and seller's surpluses is

$$\frac{u(y)-a}{-y+a} = u'(y)\frac{\delta_s}{\delta_b}.$$

The buyer's share of the surplus depends on two forces. The first is the relative breakdown hazards faced by the two players, summarized by δ_s/δ_b. The second is the buyer's marginal utility of consumption, $u'(y)$, which captures the tightness of the liquidity constraint. A tighter liquidity constraint, i.e., higher $u'(y)$, therefore increases the buyer's bargaining share. The buyer's endogenous share in the total surplus of the match is

$$\Theta = \frac{u'(y)\delta_s}{\delta_b + u'(y)\delta_s}.$$

Recall from the main text that

$$\Theta(y) \equiv \frac{\theta u'(y)}{\theta u'(y) + 1 - \theta}.$$

Hence, the solution to the alternating-offer bargaining game corresponds to the generalized Nash solution used in the text when $\theta = \delta_s/(\delta_s + \delta_b)$.

B3. A finite-horizon bargaining game

We now consider an extensive-form, finite-horizon bargaining game with at most two rounds of offers and counteroffers. While this game does not provide foundations for the Nash solution, there are nevertheless two motivations for introducing it. First, it is a simple alternative to the infinite-horizon, alternating-offer framework that might be more naturally interpreted as occurring within a single period. Second, it allows us to demonstrate that the non-monotonicity of the Nash solution is a robust feature of bargaining games with liquidity constraints.

The game The timing of the game is as follows. The buyer makes the first offer. If the offer is rejected, the second proposer is chosen randomly: the buyer with probability θ and the seller with probability $1-\theta$. The offer is then accepted or rejected and the game ends. The notion of equilibrium is subgame perfection and the game is solved by backward induction.

Second round of the game In the second round of the game, if the buyer is chosen to make a take-it-or-leave-it offer, then she proposes

$$\begin{aligned} y_2^b(a) &= \min\{y^*, a\} \\ p_2^b(a) &= \min\{y^*, a\}. \end{aligned}$$

The buyer extracts the whole surplus by equalizing the payment to the seller's production cost. If the seller is chosen to make the offer,

$$\begin{aligned} y_2^s(a) &= \min\{y^*, u^{-1}(a)\} \\ p_2^s(a) &= \min\{u(y^*), a\}. \end{aligned}$$

Now the seller extracts the whole surplus by equalizing the payment with the buyer's utility.

First round of the game In the first round, the buyer makes an offer that makes the seller indifferent between accepting or rejecting. If she rejects, she will be able to make a take-it-or-leave-it offer with probability $1-\theta$. Hence, the first-round offer is a solution to the following problem:

$$\max_{y,p\leq a} \{u(y) - p\} \quad \text{s.t. } p - y = (1-\theta)\left[p_2^s(a) - y_2^s(a)\right].$$

If $a \geq u(y^*)$, then $y = y^*$ and $p = p^* \equiv \theta y^* + (1-\theta)u(y^*)$. If $a < u(y^*)$ but $p \leq a$ is slack, the optimal offer is

$$\begin{aligned} y_1^b(a) &= y^* \\ p_1^b(a) &= y^* + (1-\theta)\left[a - u^{-1}(a)\right]. \end{aligned}$$

The liquidity constraint is slack if $y^* \leq \theta a + (1-\theta)u^{-1}(a)$, which can be reexpressed as $a \geq \hat{a}$ where $y^* = \theta\hat{a} + (1-\theta)u^{-1}(\hat{a})$. Finally, if $a < \hat{a}$ and $p \leq a$ binds, then the solution is:

$$\begin{aligned} y_1^b(a) &= \theta a + (1-\theta)u^{-1}(a) \\ p_1^b(a) &= a. \end{aligned}$$

The buyer's surplus From the solution to the bargaining game above, the buyer's surplus is given by:

$$\Delta^b(a) = \begin{cases} \theta\left[u(y^*) - y^*\right] \\ u(y^*) - y^* - (1-\theta)\left[a - u^{-1}(a)\right] \\ u\left[y_1^b(a)\right] - a \end{cases} \quad \text{if } a \begin{cases} \geq u(y^*) \\ \in [\hat{a}, u(y^*)) \\ < \hat{a} \end{cases}$$

The derivative of this surplus function is:

$$\Delta^{b\prime}(a) = \begin{cases} 0 \\ -(1-\theta)\left[1 - \frac{1}{u'[u^{-1}(a)]}\right] \\ u'\left[y_1^b(a)\right] \times \left\{\theta + \frac{1-\theta}{u'[u^{-1}(a)]}\right\} - 1 \end{cases} \quad \text{if} \quad a \begin{cases} \geq u(y^*) \\ \in [\hat{a}, u(y^*)) \\ < \hat{a} \end{cases}$$

For all $a \in [\hat{a}, u(y^*))$ and $\theta < 1$, $\Delta^{b\prime}(a) < 0$. This proves that the buyer's surplus function is non-monotone. The intuition for why $\Delta^b(a)$ is decreasing when $a \in [\hat{a}, u(y^*))$ is different from the one of the infinite-horizon game. If $a > \hat{a}$, the buyer's liquidity constraint is slack. By increasing her asset holdings above $\hat{a}$, the buyer raises the seller's expected payoff following a rejection, as the seller can then demand a larger payment in exchange for the same output. A similar logic applies when asset holdings are close to, but below, $\hat{a}$, in which case the liquidity constraint binds.

B4. Welfare measures and the optimal monetary policy

An alternative welfare criterion to the one used in the main text is the Golden Rule criterion. Let $\mathcal{U}$ denote the expected lifetime utility of a member of a future generation in a stationary, symmetric allocation. Let c_o and h_o denote, respectively, the old agent's expected CM consumption and work, averaging over the possible outcomes of DM trade. Then,

$$\mathcal{U} = U(c) - h + \beta \left\{ \alpha\sigma \left[u(y) - y \right] + c_o - h_o \right\}.$$

We maintain Assumption 3 and $i_m \in [0, \bar{\iota})$, so that the lump-sum tax is feasible for all old agents. The budget equations for the young and the old are

$$\begin{aligned} h &= c + qA \\ h_o + qA(1 + i_m) &= c_o + \tau. \end{aligned}$$

From the budget constraint of the government, $\tau = qAi_m$. Hence, the budget constraint of the old can be reexpressed as

$$h_o + qA = c_o.$$

Substitute h and h_o from the budget identities into the objective to obtain:

$$\mathcal{U} = U(c) - c - Aq + \beta \left\{ \alpha\sigma \left[u(y) - y \right] + Aq \right\}.$$

When young, the agent enjoys utility $U(c) - c$ from consumption and incurs disutility Aq to acquire the asset. When old, the expected surplus of the agent in the DM is $\alpha\sigma \left[u(y) - y \right]$. The old sell the asset in the CM in exchange for $Aq(1 + i_m)$ units of good and pays taxes, $\tau = Aqi_m$ so that her CM surplus is equal to Aq. The welfare of future generations can then be rewritten as

$$\mathcal{U} = U(c) - c - (1 - \beta)Aq + \beta\alpha\sigma \left[u(y) - y \right].$$

This welfare measure differs from the utilitarian measure used in the main text when $\beta < 1$.

Suppose $\beta = 1$ so that the two measures coincide, $\mathcal{U} = U(c) - c + \alpha\sigma \left[u(y) - y \right]$. The Friedman rule corresponds to $R = 1$. The threshold $\hat{R}$ solves

$$\begin{aligned} & U(c^*) - c^* + \max_{a \in [0, p^*]} \left\{ \left(\frac{\hat{R} - 1}{\hat{R}} \right) a + \alpha\sigma\Delta^b(a) \right\} \\ = & \max_{c \leq \bar{h}_1 - p^*/\hat{R}} \left\{ U(c) - \hat{R}c \right\} + \left(\hat{R} - 1 \right) \bar{h}_1 + \alpha\sigma\Delta^b(p^*). \end{aligned}$$

Recall that $\hat{R}$ is the threshold that makes agents indifferent between choosing the proposed allocation and deviating by choosing real money balances greater than p^*. It is easy to check that $\hat{R} > 1$. Indeed, if $\hat{R} = 1$ then the left side dominates the right side because

$$\max_{a \in [0,p^*]} \left\{ \alpha\sigma\Delta^b(a) \right\} = \alpha\sigma\Delta^b(\tilde{a}) > \alpha\sigma\Delta^b(p^*).$$

As $\hat{R} \to +\infty$, the left side is bounded while the right side tends to $+\infty$. For all $R \in \left(1, \hat{R}\right)$, $\mathcal{U} = U(c^*) - c^* + \alpha\sigma\left[u(y) - y\right]$ where $y < y^*$ is an increasing function of R. Thus, the welfare measure $\mathcal{U}$ increases with R for all $R \in \left(1, \hat{R}\right)$. Thus, even under the Golden Rule criterion, the optimal monetary policy requires a rate of return on money strictly greater than that implied by the Friedman rule.

B5. Concealable and segregable asset holdings

In accordance with the literature (e.g., Lagos and Wright, 2003, 2005), our model assumes that the buyer's asset holdings are observable in the DM and that the buyer cannot commit not to use them in a match. We now relax this assumption by allowing buyers, in some meetings, to conceal and segregate part of their liquid asset holdings before bargaining. We show that perfect segregability eliminates negative liquidity premia. However, the benchmark result is restored whenever there is a positive probability, however small, that assets cannot be concealed or segregated.

In practice, the ability to conceal and segregate liquid assets is likely to vary across transactions, depending on the types of assets being traded and the identity of the counterparties. Some assets can be placed in separate accounts or otherwise made unavailable for payment, while others are known to counterparties or difficult to withhold once a transaction takes place. Thus, perfect segregability in all transactions is a polar case. Our formulation accommodates different degrees of asset segregability. For our results to go through, it is only required that assets cannot be perfectly concealed and segregated in all meetings.

Formally, we now assume that there are two types of meetings in the decentralized market. There is a fraction $\lambda \in [0,1]$ of meetings where the buyer's asset holdings are common knowledge in the match and the buyer cannot commit not to spend her liquid assets. In the remaining fraction of matches, the buyer has the option to conceal and segregate some of her liquid assets when the match is formed and before the negotiation takes place. The segregated assets cannot be used for payments in the DM. The case studied in the paper corresponds to $\lambda = 1$. We will show that our results go through for all $\lambda > 0$.

We now define the expected surplus of a buyer in the DM as

$$\bar{\Delta}^b(a) = \lambda \Delta^b(a) + (1-\lambda)\tilde{\Delta}^b(a). \tag{68}$$

With probability λ the buyer cannot conceal assets and the trade surplus is $\Delta^b(a)$ as defined in the main text. With probability $1-\lambda$ the buyer has the technology to segregate some of her assets. Her surplus is

$$\tilde{\Delta}^b(a) = \begin{cases} \Delta^b(a) \text{ if } a \leq \tilde{a} \\ \Delta^b(\tilde{a}) \text{ if } a > \tilde{a}. \end{cases} \tag{69}$$

Recall that $\Delta^{b\prime}(\tilde{a}) = 0$. Thus, $\tilde{\Delta}^b(a)$ is differentiable everywhere. Substituting the expression for $\tilde{\Delta}^b(a)$ from (69) into (68), we obtain:

$$\bar{\Delta}^b(a) = \begin{cases} \Delta^b(a) & \text{if } a \leq \tilde{a} \\ \lambda \Delta^b(a) + (1-\lambda)\Delta^b(\tilde{a}) & \text{if } a > \tilde{a}. \end{cases}$$

It can be checked that $\bar{\Delta}^b(a)$ inherits the key properties from $\Delta^b(a)$. In particular, using that $p^* > \tilde{a}$,

$$\bar{\Delta}^{b\prime}(p^{*-}) = \lambda \Delta^{b\prime}(p^{*-}) < \bar{\Delta}^{b\prime}(p^{*+}) = 0 \text{ for all } \lambda > 0.$$

Provided that there is a positive fraction of matches in which buyers cannot conceal or segregate their holdings, no matter how small this fraction is, the expected trade surplus of the buyer is non-monotone in a and exhibits a non-differentiability and non-convexity at $a = p^*$.

The DM value function of the buyer is now given by

$$V(a) = \alpha\sigma\bar{\Delta}^b(a) + \alpha\sigma\Delta^s + a.$$

The optimal choice of asset holdings of a young agent is given by

$$\max_{a \in \left[0, R\bar{h}_1\right]} \Gamma(a)$$

where

$$\Gamma(a) \equiv \max_{c \leq \bar{h}_1 - a/R} \left\{ U(c) - c + \beta \left[-\left(\frac{1-\beta R}{\beta R} \right) a + \alpha\sigma\bar{\Delta}^b(a) \right] \right\} \quad \forall a \in \left[0, R\bar{h}_1\right].$$

Here, for simplicity, we assume that $\Delta^b(a)$ is strictly concave for all $a \in (0, p^*)$.

We start with the polar case where $\lambda = 0$. In this case, $\bar{\Delta}^b(a)$ is concave and differentiable everywhere with $\bar{\Delta}^{b\prime}(p^*) = 0$. The following proposition establishes that in that case, the equilibrium rate of return of the asset, R^*, is always such that $R^* \leq R^0$, i.e., the liquidity premium is always nonnegative.

Proposition 11 ***(The asset is fully segregable.)*** *Suppose $\lambda = 0$. For all $A < A_0$, $R^* < R^0 = 1 + \rho$. For all $A \geq A_0$, $R^* = R^0$.*

Proof. Consider first equilibria where the constraint $c \leq \bar{h}_1 - a/R$ does not bind. The asset demand is given by the first-order condition of the problem above,

$$\frac{1-\beta R}{\beta R} = \alpha\sigma\bar{\Delta}^{b\prime}(a) = \alpha\sigma \max\left\{\Delta^{b\prime}(a), 0\right\}.$$

By market clearing,

$$a = A\left(\frac{dR}{R-1}\right).$$

Combining these two equations, the equilibrium R, denoted R^*, solves

$$\frac{1-\beta R}{\beta R} = \alpha\sigma \max\left\{\Delta^{b\prime}\left[A\left(\frac{dR}{R-1}\right)\right], 0\right\}.$$

Using the definition of A_0 as

$$A_0 \equiv \frac{(1-\beta)\tilde{a}}{d},$$

it follows that for all $A < A_0$, $R^* < R^0 = \beta^{-1}$ and $\bar{\Delta}^{b\prime}(a) > 0$. For all $A > A_0$, $R^* = R^0 = \beta^{-1}$ and $\bar{\Delta}^{b\prime}(a) = 0$. The constraint $c \leq \bar{h}_1 - a/R$ binds when

$$c^* > \bar{h}_1 - \frac{a}{R^*} = \bar{h}_1 - \frac{Ad}{\rho}.$$

Equivalently,

$$A > \breve{A}_1 \equiv \frac{\rho\left(\bar{h}_1 - c^*\right)}{d}.$$

In that case $c = \bar{h}_1 - a/R$ and the first-order condition together with the market-clearing condition give

$$U'\left(\bar{h}_1 - \frac{Ad}{R-1}\right) = \beta R.$$

This corresponds to the definition of R^0 when $A > \breve{A}_1$. Thus, $R^* = R^0$. ■

We now turn to the general case where $\lambda > 0$, i.e., liquid assets are not perfectly segregable.

Proposition 12 ***(The asset is partially segregable.)*** *Suppose $\lambda > 0$. For all $A < A_0$, $R^* < R^0 = 1 + \rho$. For A in the right neighborhood of A_0, $R^* > R^0 = 1 + \rho$ is the solution to*

$$\frac{1 - \beta R^*}{\beta R^*} = \alpha\sigma\lambda\Delta^{b\prime}\left(\frac{AdR^*}{R^* - 1}\right). \tag{70}$$

As λ increases, R^* increases.

Proof. The case $A < A_0$ is the same as in Proposition 11. Consider A in the right neighborhood of A_0. By the same reasoning as in the main text, we can focus on symmetric equilibria where the constraint $c \leq \bar{h}_1 - a/R$ does not bind. The asset demand is given by the first-order condition of the portfolio problem,

$$\frac{1 - \beta R}{\beta R} = \alpha\sigma\bar{\Delta}^{b\prime}(a) = \alpha\sigma\lambda\Delta^{b\prime}(a).$$

By market clearing,

$$a = A\left(\frac{dR}{R-1}\right).$$

Combining these two equations gives (70). When $A = A_0$, $R^* = 1 + \rho$. Since $A > A_0$, and using that the left side of (70) is decreasing in R^* while the right side is increasing, $R^* > 1 + \rho$. To understand why the right side is increasing, recall that $\Delta^{b\prime}$ is decreasing and

the argument, $AdR^*/(R^*-1)$, is also decreasing in R^*. As λ increases, the right side of (70) decreases. To maintain the equality, R^* increases. ■

Proposition 12 establishes that negative liquidity premia exist for a range of asset supplies provided that $\lambda > 0$, i.e., liquid assets are not perfectly segregable. Hence, an arbitrarily small fraction of meetings in which asset holdings cannot be concealed or segregated is sufficient to generate negative liquidity premia. As segregability diminishes, R^* increases and liquidity premia become more negative.

B6. Collateralized borrowing with rehypothecation

We extend the model by allowing young agents to borrow and lend against collateral in the CM. We assume that collateral can be rehypothecated: an agent who receives an asset as collateral can use it as a means of payment in the subsequent DM. We show that collateralized borrowing and lending are neutral for equilibrium allocations and asset prices.

Let $b \geq 0$ denote the face value of collateralized borrowing and let $\ell \geq 0$ denote the face value of collateralized lending (in terms of the CM good). We represent the balance sheet of an agent as follows:

Assets	Liabilities and net worth
a	b
ℓ	$x \equiv a + \ell - b$

Due to lack of commitment, we assume one-for-one collateralization, so that an agent who borrows b pledges assets worth b, while an agent who lends ℓ receives assets worth ℓ as collateral. Let R^b denote the gross real rate of return on collateralized loans. Collateral received through lending can itself be repledged to secure additional CM borrowing. Hence, collateralized borrowing is subject to

$$b \leq a + \ell,$$

or, equivalently $x \geq 0$. This restriction simply requires that the value of collateral pledged by an agent does not exceed the value of the assets under her control, including collateralized loans.

The problem of a young agent becomes

$$W_y = \max_{c,h,a,b,\ell} \{U(c) - h + \beta V(a, b, \ell)\}$$

subject to

$$c + \frac{a}{R} + \frac{\ell}{R^b} = h + \frac{b}{R^b}, \qquad 0 \leq h \leq \bar{h}_1,$$

and $a, b, \ell, c, h \geq 0$ and $b \leq a + \ell$. The value function of the agent in the DM is given by

$$V(a, b, \ell) = \alpha\sigma\Delta^b(a - b + \ell) + \alpha\sigma\Delta^s + a - b + \ell.$$

The argument of Δ^b reflects rehypothecation. An agent who purchases a units of the asset but borrows b against collateral relinquishes control of b units during the DM. Conversely, an agent who lends ℓ receives ℓ units of collateral that can be used in the DM. Hence, the agent enters the DM with $a - b + \ell$. The problem of the young agent can be written as

$$W_y = \max_{a,b,\ell \geq 0,\ b \leq a+\ell} \{\Gamma(a, b, \ell) + \beta\alpha\sigma\Delta^s\},$$

where

$$\Gamma(a,b,\ell) \equiv \max_{0\le c\le \bar{h}_1 - a/R + (b-\ell)/R^b} \{U(c) - c + \beta\left[-\left(\frac{1-\beta R}{\beta R}\right)a - \left(\frac{1-\beta R^b}{\beta R^b}\right)(\ell - b) + \alpha\sigma\Delta^b(a+\ell-b)\right]\}.$$

Collateralized borrowing and lending therefore affect an agent's liquidity position only through her net holdings,

$$x = a - b + \ell.$$

Notice that the collateral feasibility condition $b \le a+\ell$ is equivalent to $x \ge 0$. Thus, allowing collateral received through lending to be pledged again does not introduce any additional restriction once the portfolio problem is expressed in terms of net liquid holdings.

We next characterize the equilibrium return on collateralized loans. Suppose first that $R^b < R$. An agent can increase a and b by the same amount, thereby leaving $x = a - b + \ell$ unchanged. This transaction leaves the agent's DM liquidity and future net wealth unchanged, while relaxing her current CM budget constraint because $1/R^b > 1/R$. Hence, the agent can increase current consumption without changing any future payoff, which is an arbitrage opportunity.

Conversely, suppose that $R^b > R$ and that an agent has a positive net borrowing position, $b - \ell > 0$. We say that collateralized borrowing and lending are active if some agents have positive net borrowing positions. The agent can reduce a and b by the same amount, leaving her DM liquidity and future net wealth unchanged while again relaxing her current CM budget constraint because $1/R > 1/R^b$. Hence, no agent would hold a positive net borrowing position. Since positive net lending requires positive net borrowing by market clearing, an equilibrium with active collateralized borrowing and lending must satisfy

$$R^b = R.$$

When $R^b = R$, the young agent's CM budget constraint becomes

$$c + \frac{a - b + \ell}{R} = h,$$

or, using $x = a - b + \ell$,

$$c + \frac{x}{R} = h.$$

Moreover,

$$-\left(\frac{1-\beta R}{\beta R}\right)a + \left(\frac{1-\beta R}{\beta R}\right)(b-\ell) = -\left(\frac{1-\beta R}{\beta R}\right)x.$$

Consequently, the young agent's problem reduces to

$$W_y = \max_{x \geq 0} \left\{ \Gamma(x) + \beta\alpha\sigma\Delta^s \right\},$$

where

$$\Gamma(x) \equiv \max_{c \leq \bar{h}_1 - x/R} \left\{ U(c) - c + \beta \left[- \left(\frac{1 - \beta R}{\beta R} \right) x + \alpha\sigma\Delta^b(x) \right] \right\}.$$

This is identical to the portfolio problem in the economy without collateralized borrowing, with x replacing a.

To complete the argument, let i index young agents. Market clearing in the collateralized loan market requires

$$\int b_i \, di = \int \ell_i \, di.$$

Therefore, aggregate liquid holdings in the DM satisfy

$$\begin{aligned} \int x_i \, di &= \int (a_i - b_i + \ell_i) \, di \\ &= \int a_i \, di. \end{aligned}$$

Thus the market clearing condition for the asset can be written as:

$$\int x_i \, di = \int a_i \, di = A(q + d) = Ad \left(\frac{R}{R - 1} \right).$$

From the problem above, for a given R,

$$\int x_i \, di \in \text{conv}[\text{A}^*(R)],$$

where $\text{A}^*(R)$ is the individual asset demand correspondence in the economy without collateralized borrowing and lending and is also the set of solutions for x in the problem above. Hence, market clearing requires

$$Ad \left(\frac{R}{R - 1} \right) \in \text{conv}[\text{A}^*(R)].$$

Thus, the equilibrium R coincides with the one in the economy without collateralized borrowing and lending. Rehypothecation can generate arbitrarily large gross positions in collateralized borrowing and lending, but it does not change the aggregate quantity of liquid assets available in the DM. Collateralized borrowing therefore does not expand agents' effective arbitrage capacity and leaves the equilibrium characterization of liquidity premia unchanged.

Proposition 13 (Neutrality of collateralized borrowing with rehypothecation) *Suppose young agents can borrow and lend against collateral and collateral can be rehypothecated, including by pledging collateral received through lending to secure additional borrowing. In any equilibrium with active collateralized borrowing and lending,* $R^b = R$. *Moreover, the equilibrium allocation and asset return coincide with those in the economy without collateralized borrowing and lending.*

B7. Nested acceptability

In the multiple-asset economy studied in Section 5, different liquid assets are accepted in distinct types of meetings. This assumption makes the equilibrium characterization tractable in the presence of the non-convexity generated by generalized Nash bargaining. We now consider an alternative environment in which liquid assets compete directly as means of payment.

There are three short-lived assets, indexed by $j \in \{0, 1, 2\}$. Asset 0 is illiquid. Asset 2 is more liquid than asset 1 in the following sense. With probability α_1, both assets 1 and 2 are accepted as means of payment. With probability $\alpha_2 - \alpha_1$, where $0 < \alpha_1 < \alpha_2 \leq 1$, only asset 2 is accepted. Thus, asset 2 can be used in every meeting in which asset 1 can be used, as well as in an additional set of meetings.

We restrict attention to symmetric equilibria in which the labor supply constraint of the young is slack. Because the portfolio problem is generally non-convex, we use only first-order necessary conditions.

Let (a_1, a_2) denote the holdings of the two liquid assets. In meetings in which both assets are accepted, the buyer can use total liquid balances $a_1 + a_2$. In meetings in which only asset 2 is accepted, the buyer can use a_2. Hence, the buyer-surplus component of the young agent's objective is

$$\beta\sigma \left\{ \alpha_1 \Delta^b(a_1 + a_2) + (\alpha_2 - \alpha_1)\Delta^b(a_2) \right\}.$$

Since the labor supply constraint is slack, $c = c^*$. The first-order necessary conditions for an interior portfolio combined with market clearing in a symmetric equilibrium, $a_j = A_j$ for $j \in \{1, 2\}$, give us the following asset prices:

$$\begin{aligned} q_0 &= \beta \\ q_1 &= \beta \left[1 + \alpha_1 \sigma \Delta^{b\prime}(A_1 + A_2) \right] \\ q_2 &= \beta \left[1 + \alpha_1 \sigma \Delta^{b\prime}(A_1 + A_2) + (\alpha_2 - \alpha_1)\sigma \Delta^{b\prime}(A_2) \right]. \end{aligned}$$

Taking the difference between the last two equations, the asset price premium for the most liquid asset is

$$q_2 - q_1 = \beta(\alpha_2 - \alpha_1)\sigma \Delta^{b\prime}(A_2).$$

The price difference between the two liquid assets depends only on the marginal value of the additional liquidity services provided by asset 2 in the meetings in which asset 1 is not accepted.

Proposition 14 (Nested acceptability) *Consider a symmetric equilibrium with a slack*

labor supply constraint and $A_1 + A_2 < p^*$ in which the first-order necessary conditions above hold.

1. *If $A_1 + A_2 < \widetilde{a}$, then*

$$q_2 > q_1 > q_0 = \beta.$$

   Hence, liquidity premia are positive and increase with acceptability.

2. *Suppose $A_2 < \widetilde{a} < A_1 + A_2$. Then*

$$q_1 < q_0 = \beta,$$

   while

$$q_2 > q_0$$

   if

$$(\alpha_2 - \alpha_1)\Delta^{b\prime}(A_2) > -\alpha_1\Delta^{b\prime}(A_1 + A_2).$$

   Hence, positive and negative liquidity premia can coexist even though the more liquid asset can be used in every meeting in which the less liquid asset can be used.

3. *If $A_2 > \widetilde{a}$, then*

$$q_2 < q_1 < q_0 = \beta.$$

   Hence, the liquidity-based yield curve is inverted: the more liquid asset trades at a lower price and earns a higher return.

The proposition shows that the cross-sectional mechanism emphasized in the main text does not rely on exclusive segmentation across meetings. When liquidity is scarce at the margin, greater acceptability raises an asset's nonpecuniary value and therefore its price. When liquidity is abundant at the margin, the opposite occurs. In particular, once $\Delta^{b\prime}(A_2) < 0$, the additional set of meetings in which asset 2 can be used lowers rather than raises its marginal nonpecuniary return relative to asset 1. Thus, direct substitution between liquid assets in decentralized trade does not eliminate the mechanism generating negative liquidity premia or the possibility of an inverted liquidity-based yield curve.

B8. Sampling uncertainty in the calibration

The two empirical moments used to calibrate B and $\alpha\sigma$—the level of the convenience yield and its semi-elasticity with respect to Treasury supply—are estimated with sampling uncertainty. To assess its implications for our quantitative results, we follow the bootstrap procedure of Jiang et al. (2025). We conduct 2,000 bootstrap replications, resampling the quarterly observations using circular blocks of eight consecutive quarters, as in Jiang et al. (2025), and re-estimate the two calibration moments for each bootstrap sample. For each admissible set of moments, we recalibrate B and $\alpha\sigma$, holding all other parameters at their benchmark values, and compute the corresponding equilibrium convenience-yield curve.

The confidence band in Figure 8 is constructed pointwise. At each value of A^1, we take the 5th and 95th percentiles of the convenience yields generated by the bootstrap calibrations. Each calibration is solved using the full equilibrium model, allowing equilibrium regimes and their transition points to vary across bootstrap samples. The resulting band therefore captures sampling uncertainty in the empirical moments used to discipline the calibration, rather than the dispersion of observed convenience yields around the model prediction.

The main qualitative prediction is robust to this uncertainty. In the benchmark calibration, the convenience yield crosses zero at $A^1 = 6.01$ percent of GDP. Across bootstrap calibrations, the median zero crossing is 6.04 percent of GDP, with a 90 percent interval of $[5.36, 7.87]$ percent of GDP. Thus, sampling uncertainty about the calibration targets does not materially affect the prediction that convenience yields become negative when the supply of liquid assets is sufficiently large.